

Direct Kinematic-Mass Target Extension of the Neutrino Standing-Carrier Space

Absolute-scale lift fibres under the KATRIN KNM1–5 endpoint constraint

PCR-5D1

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Principal theorem

Let $\widehat{T}_K^{\text{BB}}$ be the target-complete direct-kinematic realization object consisting of the retained oscillation target, the checksum-locked KNM1–5 empirical packet, and the explicitly declared enriched source comparison class **SOE-D1-1**. For

$$j_\beta^{\text{BB}} : T_{\text{osc}} \hookrightarrow \widehat{T}_K^{\text{BB}}, \quad \mathcal{L}_{\beta, \text{BB}}(q) = \mathcal{N}(\widehat{T}_K^{\text{BB}}) \underset{\mathcal{N}(T_{\text{osc}})}{\overset{h}{\times}} \mathcal{N}(T_{\text{osc}})[q],$$

the source-authorization theorem assigns to every finite source-coherent PCR-3 carrier hypergraph H a colored hyperprocess envelope $\text{BB}(H)$. The envelope preserves irreducible higher incidence, admits deletion on the current enriched source, and has exact base reduct

$$U_\beta^{\text{BB}}(\text{BB}(H)) = H.$$

Complete downstream evaluation gives

$$\Phi_4^{\text{ret}}(\text{BB}(H)) = \Phi_4(H), \quad \Xi_{\text{osc}}^{\text{ret}}(\text{BB}(H)) = \Xi_{\text{osc}}(H),$$

including the frozen no-repair deletion field. Because the hyperprocess presentation is beta-endpoint neutral before the D1 metric law is attached, the one-support family

$$K_{\beta, t} = t\mathbf{1}, \quad \mu_{\beta, t} = \delta_t, \quad 0 \leq t < (0.45 \text{ eV})^2,$$

realizes the exact one-parameter KNM1–5 response. Choosing one framed provenance presentation in each base class yields

$$\boxed{\text{im } \rho_{\beta, \text{BB}} = Q_{\text{osc}}^{\text{SCIC}}}, \quad \boxed{|\pi_0 \mathcal{L}_{\beta, \text{BB}}(q)| \geq \mathfrak{c} \quad \forall q \in Q_{\text{osc}}^{\text{SCIC}}}.$$

Thus the direct-kinematic target eliminates no oscillation-level SCIC class and refines every one into at least continuum many standing-distinct enriched realizations. The section is choice-dependent; no canonical representative-independent section is claimed.

Target and comparison boundary

The empirical component is the KATRIN KNM1–5 tritium beta-decay analysis, including its released records, response inputs, covariance structure, calibration relations, likelihood, and acceptance semantics. The target-complete object also includes the versioned source comparison class **SOE-D1-1**. No cosmological mass-sum likelihood, Majorana/LNV record, direct heavy-state target, or $B - L$ condition is imported, and the PCR-5B base quotient is unchanged by the enriched source extension.

Abstract

PCR-5B determines the standing-carrier quotient of the neutrino-oscillation target as a mixed discrete, continuous, functional, and categorical moduli object, while PCR-5C supplies the general calculus of target restriction, lift fibres, elimination, and splitting. This paper applies that calculus to the direct tritium beta-decay endpoint target defined by the KATRIN KNM1–5 analysis.

The empirical target incorporates the 259-day KATRIN result, the released campaign records and model inputs, the molecular final-state and apparatus-response structure, the nuisance and covariance architecture, and the published 90% confidence bound $m_\beta < 0.45$ eV. A negative fitted value of m_β^2 is retained as a statistical continuation coordinate and is never interpreted as negative physical spectral support.

The complete target also fixes its source comparison class. Frozen PCR-3 does not contain a universal operation that turns an arbitrary source-coherent carrier hypergraph into an endpoint-neutral process presentation. The direct-kinematic target therefore declares the conservative enriched source extension **SOE-D1-1**. Its colored hyperprocess envelope preserves primitive incidences of every arity, source equations, boundary and matching data, higher coherence, and current-source deletion. The enriched object has an exact reduct to the original PCR-3 source point, and only after source formation does the PCR-4/PCR-5B evaluator establish equality of the retained oscillation trace and all seven standing channels.

The endpoint-neutral hyperprocess presentation admits the exact one-support KATRIN family $\mu_{\beta,t} = \delta_t$ for $0 \leq t < (0.45 \text{ eV})^2$. Selecting one framed source presentation in each base class gives a right inverse of the restriction map and an injection of a real interval into every lift fibre. Consequently

$$\text{im } \rho_{\beta,\text{BB}} = Q_{\text{osc}}^{\text{SCIC}}, \quad |\pi_0 \mathcal{L}_{\beta,\text{BB}}(q)| \geq \mathfrak{c} \quad \forall q \in Q_{\text{osc}}^{\text{SCIC}}.$$

Intrinsic discrete, kinked, functional, continuous, hidden, composite, solitonic, environmental, boundary, and mixed endpoint realizations remain additional source-faithful components or typed complement strata; the universal section does not overwrite them.

Interpretive boundary

The theorem is relative to the complete target $\hat{T}_K^{\text{BB}}$, which includes both the KNM1–5 empirical packet and the declared **SOE-D1-1** source comparison class. The public-data audit establishes target identity and field completeness but does not reproduce the collaboration fit. AASC constrains and closes inadmissible branches; it does not generate source candidates or infer an occupant from an empty candidate fragment.

Keywords. AASC; neutrino mass; KATRIN; tritium beta decay; direct kinematic mass; standing carrier; SCIC; target extension; hyperprocess; lift fibre; spectral measure; absolute scale.

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1 Direct kinematic extension of the oscillation carrier space

1.1 The target-extension problem

PCR-5B closes the standing-carrier quotient of the neutrino-oscillation target, including the LAW, LOCAL, HID, COMP, SOL, and HYB charts, their pairwise and higher intersections, the typed failure boundary, and the framed source provenance associated with every standing core [9]. PCR-5C proves that a lawful target extension induces a restriction functor on standing cores, connected-class lift groupoids, preservation of retained base distinctions, and exact classwise elimination or splitting rules [10].

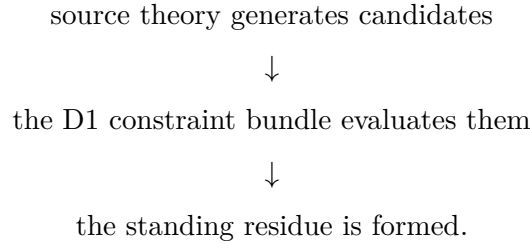
The present paper enlarges the target by the direct electron-boundary kinematic record fixed by the KATRIN KNM1–5 analysis. Its question is global rather than mechanism-specific:

For each connected class $q \in Q_{\text{osc}}^{\text{SCIC}}$, what is the exact standing residue of all source-lawful direct-kinematic completions after the KATRIN, source, boundary, response, continuation, matching, failure, and restriction constraints are imposed simultaneously?

A source presentation is not the same object as a standing-carrier class. One presentation may fail the enriched target while another presentation of the same base standing core survives. Elimination of a base class therefore requires exhaustion of the entire framed provenance fibre and every lawful D1 completion over it.

1.2 Constraint-native order of proof

The argument observes the order



The null-regime theorem is used only as a same-type closure theorem: total vacuity cannot occupy the positive side of an admissibility-bearing fixed-domain problem, but null exclusion does not supply a concrete occupant [18]. Likewise, realized-persistence non-vacuity applies only after standing-bearing persistence is already realized and does not convert an empty candidate fragment into existence [14]. The positive D1 residue is therefore obtained from an independently defined source class followed by simultaneous evaluation.

1.3 The complete D1 target

The complete object used throughout the paper is

$$\hat{T}_K^{\text{BB}} = (T_{\text{osc}}, T_K^{\text{emp}}, \mathbb{S}_\beta^{\text{BB}}, \text{SOE-D1-1}, \mathcal{C}_\beta^{\text{BB}}, \mathcal{A}_\beta^{\text{BB}}, \mathcal{W}_\beta, \mathcal{U}_\beta, \Xi_\beta^{\text{BB}}, \mathcal{N}_\beta^{\text{BB}}, Q_\beta^{\text{BB}}).$$

Here T_K^{emp} is the checksum-locked KNM1–5 empirical packet. The remaining entries specify the enriched source theory, comparison class, candidate and survivor groupoids, witness and continuation structures, standing packet, core groupoid, and quotient. This distinction is essential: the strong theorem is a statement about the empirical KATRIN target together with the explicitly declared source comparison

class, not about the empirical record in isolation.

Frozen PCR-3 contains finite source-coherent hypergraphs, law/process objects, derived objects, finite hybrids, and irreducible higher incidence, but it does not assert a universal process-abstraction operation. The enriched target therefore declares **SOE-D1-1**, adding a colored hyperprocess sort, a constructor

$$H \mapsto \text{BB}(H),$$

current-source deletion functors, and an exact reduct

$$U_{\beta}^{\text{BB}}(\text{BB}(H)) = H.$$

The extension is fixed before survivor evaluation. It is not attributed to PCR-3 saturation and does not alter the frozen oscillation source ontology or quotient.

1.4 Source authorization and retained standing

For every finite, source-coherent, activity-reduced carrier hypergraph H , the hyperprocess envelope preserves every primitive ordinary incidence and every irreducible higher-arity incidence as an operation of the same arity. It retains the full parameter object, boundary, matching, representation, charge, anomaly, vacuum, continuation, gluing, and higher-coherence data. Deletion is performed on the current enriched source and satisfies

$$\text{Del}_S^{\text{BB}}(\text{BB}(H)) \cong \text{BB}(\text{Del}_S H).$$

Only after source formation does the downstream evaluator establish

$$\Phi_4^{\text{ret}}(\text{BB}(H)) = \Phi_4(H), \quad \Xi_{\text{osc}}^{\text{ret}}(\text{BB}(H)) = \Xi_{\text{osc}}(H).$$

Thus the enriched object is a distinct source presentation with the same retained oscillation standing core, not a replacement of PCR-3 source identity.

1.5 Exact endpoint completion and main result

Before the direct-kinematic law is attached, the hyperprocess presentation contains no absolute beta spectral object. The one-support family

$$K_{\beta,t} = t\mathbf{1}, \quad \mu_{\beta,t} = \delta_t, \quad 0 \leq t < (0.45 \text{ eV})^2,$$

therefore supplies the complete endpoint measure of that presentation. Substitution into the carrier-neutral beta functional produces exactly the frozen one-parameter KNM1–5 response at $m_{\beta}^2 = t$, with the same final-state, apparatus, background, covariance, nuisance, likelihood, and acceptance maps.

For each $q \in Q_{\text{osc}}^{\text{SCIC}}$, choose one framed B15 presentation H_q . The accepted family $\mathfrak{E}_{\beta,t}^{\text{BB}}(H_q)$ restricts to q , so the restriction map is onto. Different t values are separated by the admitted absolute-support channel. Hence

$$\text{im } \rho_{\beta,\text{BB}} = Q_{\text{osc}}^{\text{SCIC}}, \quad |\pi_0 \mathcal{L}_{\beta,\text{BB}}(q)| \geq \mathfrak{c} \quad \forall q \in Q_{\text{osc}}^{\text{SCIC}}.$$

The choice of framed provenance need not be canonical. A richer target may make previously standing-inert provenance visible, so the theorem claims a choice-dependent right inverse rather than representative-independent enriched identity.

1.6 Structure of the argument

The proof proceeds through five stages. First, the KNM1–5 empirical packet and the complete D1 target are fixed. Second, the electron-boundary spectral measure and the exact response/likelihood maps are defined. Third, the enriched source class and its hyperprocess envelope are shown to be source coherent, higher-incidence faithful, deletion natural, endpoint neutral, and conservative under base restriction. Fourth, the class-indexed lift fibres, parameter descent, source composition, overlap structure, and failure boundary are determined. Finally, the choice-dependent section and absolute-support separator yield the full base image and continuum lower bound in every fibre.

1.7 Scope

No cosmological mass-sum likelihood, Majorana or lepton-number-violating record, heavy-state search target, nonunitarity constraint, or $B - L$ condition is imported. The KATRIN effective mass is not identified with one microscopic eigenmass, and ordering, naturalness, familiarity, field count, provenance, and ultraviolet preference have no selecting authority. The collaboration fit is not reproduced; its published target and public numerical releases are adopted as the empirical object. Intrinsic source-derived endpoint spectra remain separate components of the enriched moduli space and are never overwritten by the universal hyperprocess section.

2 Upstream structure and the no-backflow boundary

2.1 Exact oscillation base space

Let

$$\mathfrak{A}_{\text{osc}}^{\text{exec}} \simeq \mathcal{N}_{5,\text{osc}}^{\text{exec}}, \quad Q_{\text{osc}}^{\text{SCIC}} \cong \pi_0 \mathfrak{A}_{\text{osc}}^{\text{exec}}$$

be the PCR-5B publication-final base atlas [9]. Its B15 interface supplies, for every family/skeleton chart, the exact survivor solution object, lawful source groupoid, concrete standing-core map, applicability and failure frontier, framed provenance, base separator, and pairwise/higher overlap incidence.

The base packet is

$$\Xi_{\text{osc}} = (T^{\text{face}}, R^{\text{stand}}, \Delta_{\text{fr}}^{\text{del}}, C^{\text{act}}, M^{\text{match}}, B^{\text{bdy}}, F^{\text{same}}).$$

All seven fields and their separator semantics are frozen. The KATRIN extension may add endpoint-response standing; it may not revise base membership, alter PCR-4 applicability, enlarge the frozen deletion field, or identify base classes separated by an admitted base witness.

PCR-5B also retains the complete source-provenance fibre of every concrete standing core. A D1 elimination claim therefore quantifies over every framed source presentation, not one preferred mechanism. Conversely, a single lawful survivor in that complete provenance fibre is sufficient to establish base-class survival.

2.2 Source generation and simultaneous evaluation

PCR-3 generates the source universe before target survival. Its LAW branch contains explicit source-coherent process objects on existing occupancy-bearing nodes with symbolic, continuous, functional, representation-valued, and categorical parameter domains; finite hybrids retain their complete incidence and higher-coherence data [16]. Its no-backflow theorem proves that source families, activity laws, and candidate identity are independent of downstream pass/fail, survivor, standing-packet, and expected-occupant data. PCR-3 also states a precise saturation boundary: a newly recognized relation is admitted

automatically only when it is already an operation of the frozen source theory; a genuinely new source operation or constitutive equation is a source-ontology extension and reopens the theorem at that locus.

The generic hyperprocess-envelope constructor required below is not attributed to frozen PCR-3. PCR-5D1 declares the single enriched-target source extension SOE-D1-1. It proves source coherence and higher-incidence preservation inside the declared D1 source theory, proves current-source deletion naturality, and equips every hyperprocess object with an exact base-reduct functor returning the frozen PCR-3 source input. Thus the base source ontology, target, packet, and SCIC quotient remain fixed; only the D1 enriched comparison class contains the additional explicitly typed presentation.

PCR-4 evaluates one source point against one simultaneous target ledger. A source point survives only when its source equations, target records, boundary, response, continuation, anomaly, vacuum, matching, extension, and common-cone relations all pass together [15]. No desired output may be inserted after failure, and no marginally successful pieces may be spliced into one survivor.

The present paper uses this division exactly. It constructs the D1 raw candidate universe from the PCR-3/B15 grammar and then applies a complete simultaneous D1 constraint bundle. AASC does not generate the candidate.

2.3 SCIC, AASC null closure, and realized persistence

Standing Carrier Identity Closure proves that equality of the complete standing continuation signature is the unique same-scope carrier identity; a source distinction not seen by that signature remains provenance rather than a second classifier [17]. Anchor–Tensor–Skin exhausts fixed-scope descriptive differences into target/definedness change, standing-bearing tensor consequence, or standing-inert presentation/provenance [7].

Two additional closure results are used with an explicit anti-overclaim boundary. The structural null-regime theorem proves that total vacuity cannot occupy the positive side of same-type admissibility-bearing evaluability; it is closure-theoretic rather than existential [18]. The realized-persistence theorem proves that an already realized standing-bearing regime cannot be reconstruction-null and therefore has a nonempty continuation-sensitive persistence interior; it also states that an empty evaluated fragment alone does not infer an occupant [14].

Accordingly:

AASC closes null, skin, hidden-selector, outside-but-governing, and same-act-repair cells;

candidate existence is proved separately by source construction.

The positive residue below uses the frozen source/provenance atlas without reversing its dependency order. B15 supplies an already generated framed source survivor over every standing core. PCR-5D1 applies the explicitly declared D1 constructor BB to that upstream source hypergraph and proves source coherence, current-source deletion naturality, exact retained-target evaluation, and exact base reduct before KATRIN acceptance is invoked. A standing core is never used as a source-formation argument. Null closure is not used as an existence axiom, and the manuscript does not claim that BB was silently present in frozen PCR-3 saturation.

2.4 Functorial target extension

PCR-5C associates to every target-complete neutrino target T a source-complete candidate groupoid $\mathcal{C}(T)$, exact survivor groupoid $\mathcal{A}(T)$, concrete standing-core groupoid $\mathcal{N}(T)$, and quotient $Q(T) = \pi_0 \mathcal{N}(T)$ [10].

A lawful embedding $j : T_0 \hookrightarrow T_1$ induces

$$R_j : \mathcal{N}(T_1) \rightarrow \mathcal{N}(T_0), \quad \rho_j = \pi_0(R_j),$$

and representative-independent lift fibres

$$\mathcal{L}_j(q) = \mathcal{N}(T_1) \overset{h}{\times}_{\mathcal{N}(T_0)} \mathcal{N}(T_0)[q].$$

A genuine extension can eliminate, preserve, or split a base class, but retained witnesses prohibit merger. PCR-5C does not presume universal liftability; PCR-5D1 must prove the residue over every q .

2.5 Scale, shape, and anchor discipline

Oscillation data determine mass-squared differences and response shape, not a common absolute squared-mass offset. The internal mass-character and anchor papers prove the corresponding firewall [8, 11–13]. In a standard three-state calibration branch one may write

$$m_i^2(t, o) = t + \delta_i(o),$$

where o indexes a permitted shape/orientation branch and t is a common offset. A lawful direct kinematic datum can constrain t only through its independently frozen endpoint-response relation. It cannot create the oscillation shape, choose a branch by preference, or infer Dirac/Majorana character.

2.6 Experimental source stack

The external target is based on six version-locked papers and two public data releases. The primary 259-day analysis defines the KNM1–5 result and systematic budget [3]. The beta-spectrum, response, and statistical framework is supplied by [5]; the analysis architecture by [1]; the updated final-state uncertainty treatment by [19]; shifted-analyzing-plane calibration by [2]; and source-potential broadening/shift by [4]. The public data and model inputs are frozen at Zenodo DOI 10.5281/zenodo.13644900 [6].

No empirical backflow

The direct endpoint target is formed before any carrier is evaluated. A cosmological mass sum, neutrinoless double-beta record, direct heavy-state search, altered beta-decay interaction, or later KATRIN campaign would define a successor target. None may be added after a lift begins to fail while retaining the identity of the original KNM1–5 problem.

3 The KATRIN KNM1–5 empirical target

3.1 Authoritative data identity

The frozen public data files are

File	Bytes	Official MD5
KATRIN_data_KNM1–5.json	80,751	7c9e30c35c394d87917cc56c7bf7ff53
KATRIN_inputs_KNM1–5.json	31,498	6fcb3fbd3059190caa95f37243a9593a

Their local SHA-256 values are respectively

d8c3001235d4d1b88b6b263085a3b37c11560300bc11677dc72ccfa95a819079,
6ff5958c0c6e117c8e0c21ead785a6cd6f1959ef2b4d22fe4aaadeacaafb399b.

Both files parse as UTF-8 JSON. A changed byte, checksum, unit convention, or campaign schema defines a revised target.

3.2 Campaign record

Let

$$\mathcal{D}_K = (qU_{c,k}, t_{c,k}, \tau_{c,k}^P, \mathcal{T}_c^P, N_{c,p,k}, \epsilon_{c,p,k})_{c,p,k}$$

denote the released campaign record. The campaign/configuration index is

$$c \in \{\text{KNM1}, \text{KNM2}, \text{KNM3-SAP}, \text{KNM3-NAP}, \text{KNM4-NOM}, \text{KNM4-OPT}, \text{KNM5}\}.$$

The field-level audit gives the summary in [table 1](#).

Table 1: Authoritative KNM1–5 campaign record. Count-vector width is one for aggregated configurations and fourteen for shifted/optimized detector-patch configurations.

Campaign	Points	Width	Count cells	Released counts	Live time (days)
KNM1	27	1	27	2,029,854	21.736
KNM2	28	1	28	4,314,055	28.927
KNM3-SAP	28	14	392	1,069,831	9.173
KNM3-NAP	28	1	28	1,426,505	9.339
KNM4-NOM	28	14	392	5,637,508	34.783
KNM4-OPT	25	14	350	4,582,387	17.990
KNM5	28	14	392	16,652,705	51.067
Total	192	–	1,609	35,712,845	173.015

3.3 Model-input record

The model-input object is

$$\mathcal{I}_K = (I_{\text{EM}}, I_{\rho d}, I_{\text{iso}}, I_{\text{RW}}, I_{\text{bg}}, I_{\text{loss}}, I_{\text{det}}, I_{\text{pot}}, I_{\text{drift}}).$$

It contains electromagnetic fields and retarding-potential offsets, gas density and covariance, isotopologue concentrations, rear-wall terms, background parameters, energy-loss values and covariance, angular-dependent detector efficiency, source-potential shift/broadening with correlations, and temporal potential-drift broadening. The audit indexed twenty-five parameter/correlation objects, seventy-nine field records, and nine covariance/correlation matrices.

Four source-typed observations are retained rather than normalized away: three explicitly null transmission-broadening uncertainties; signed fitted potential-drift-broadening-squared values; background pixel-group metadata not in one-to-one correspondence with values; and finite-decimal gas-density covariance asymmetry whose symmetrized matrix is positive semidefinite. None is treated as zero or silently edited.

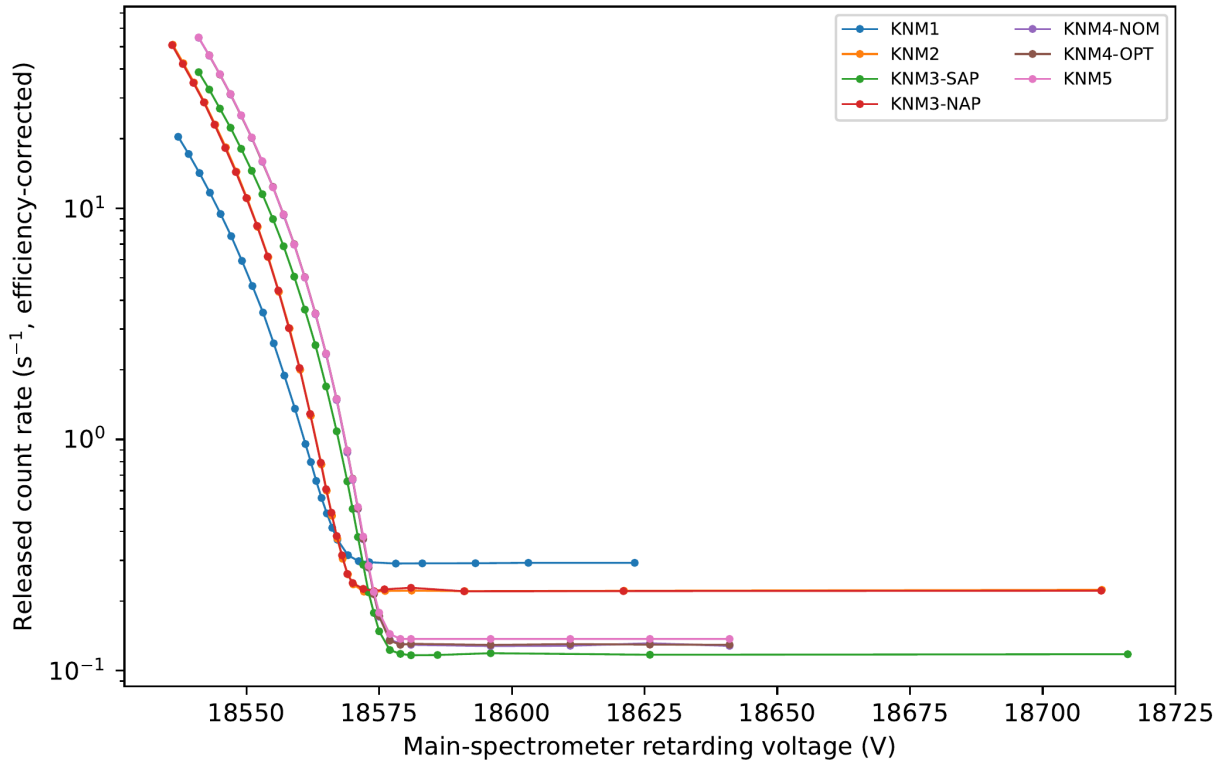


Figure 1: Efficiency-corrected released count rate versus main-spectrometer retarding voltage, calculated directly from the public campaign JSON. The plot is descriptive of the frozen record and is not a reconstruction of the collaboration fit.

3.4 Published inference record

The primary analysis combines 259 measurement days and approximately 36 million electrons in the analysis range. It reports [3]

$$\hat{m}_\beta^2 = -0.14_{-0.15}^{+0.13} \text{ eV}^2$$

and

$$m_\beta < 0.45 \text{ eV} \quad (90\% \text{ CL}).$$

The displayed uncertainty budget includes a statistical contribution of 0.108 eV^2 and systematic contributions dominated by column-density times inelastic-cross-section uncertainty, the energy-loss function, scan-step-duration-dependent background, source-potential variation, and non-Poissonian background.

3.5 Data completeness

Theorem 3.1 (Authoritative KNM1–5 data readiness). *The predicate*

$$\begin{aligned} \text{D1DataReady} \iff & \text{PrimaryPapersLocked} \wedge \text{DataJSONCached} \wedge \text{InputsJSONCached} \\ & \wedge \text{ByteCountsVerified} \wedge \text{MD5Verified} \wedge \text{JSONParsed} \\ & \wedge \text{FieldDictionaryComplete} \wedge \text{UnitsAndCovarianceTyped} \\ & \wedge \text{PaperInputProvenanceBound} \end{aligned}$$

holds for the frozen project source.

Proof. The two embedded files have the exact official byte counts and MD5 values stated above, parse successfully, and reproduce the seven campaign/configuration blocks. Every measured-data key and every model-input object is mapped to a unit, index domain, shape, and target role. All nine published covariance/correlation matrices pass dimensional, finiteness, symmetry-to-published-precision, diagonal, bounded-correlation, and positive-semidefinite checks. The source map associates every load-bearing input class with the primary or calibration paper that supplies its physical semantics. The complete machine-readable audit contains 2,383 hard checks and zero hard failures. Therefore each conjunct holds. $\square$

Remark 3.2. Theorem 3.1 closes data acquisition and structural target completeness. It does not claim an independent reproduction of the collaboration maximum-likelihood fit. The numerical standard-branch confidence interval is imported from the primary analysis, while the structural lift theorems are proved locally.

3.6 Complete target fingerprint and update boundary

The empirical component has fingerprint

$$\mathbf{f}_{\text{emp}}(T_K^{\text{emp}}) = (\text{DOI}, h_{\text{data}}, h_{\text{inputs}}, \mathcal{D}_K, \mathcal{I}_K, \text{Spec}_K, \text{Resp}_K, \mathcal{B}_K, \mathcal{L}_K, \mathcal{C}_K, \mathcal{U}_K, \mathcal{A}_K).$$

The target-complete fingerprint additionally fixes the enriched source theory and comparison class:

$$\begin{aligned} \mathbf{f}(\hat{T}_K^{\text{BB}}) &= (T_{\text{osc}}, \mathbf{f}_{\text{emp}}(T_K^{\text{emp}}), \text{SOEID}, h_{\text{SOE}}, \text{HypProc}, \text{BB}, \text{Del}^{\text{BB}}, U_\beta^{\text{BB}}, h_{\mathcal{E}_{\text{BB}}}, \text{ComparisonClassID}), \\ \text{SOEID} &= \text{SOE-D1-1}, \quad \text{ComparisonClassID} = \text{D1-KNM1-5-BB/1.0}. \end{aligned}$$

The source-extension signature, constitutive equations, current-source deletion action, reduct functor, and comparison-class identifier are therefore part of the target data rather than an implicit auxiliary

choice. A new campaign, revised public file, changed analysis window, altered source/detector interaction, different confidence construction, changed response class, or changed source comparison class defines a successor target. A redescription is lawful only when it preserves the complete fingerprint.

4 Carrier-neutral electron-boundary spectral measure

4.1 Why a spectral measure is the correct carrier-neutral variable

The standard beta-decay formula is usually written as an incoherent sum over neutrino mass eigenstates. That representation is suitable for the conventional three-state model but is too narrow for a carrier-space exhaustion theorem. The target sees the energy and weight with which electron-associated neutrino excitations are emitted. It does not initially know whether those excitations arise from elementary fields, hidden dilations, a composite sector, a continuum, or a hybrid source architecture.

Definition 4.1 (Electron-boundary spectral measure). For a same-target carrier realization x , its electron-boundary mass-squared spectral measure is a finite positive Borel measure

$$\mu_e(x) \in \text{Meas}_+([0, \infty))$$

whose mass at a Borel set A is the total beta-decay boundary weight carried by physical neutrino excitations with squared mass in A . The measure is normalized to the active beta-decay boundary weight admitted by the target. In a finite discrete realization,

$$d\mu_e(s) = \sum_{a=1}^r w_a \delta(s - m_a^2) ds, \quad w_a \geq 0.$$

A continuous or mixed realization is allowed when generated by the D1 source ontology.

The measure is target-relative. A heavy source block with no active electron-boundary incidence contributes no mass to μ_e even if it is essential elsewhere in the carrier. Conversely, an electron-coupled hidden state contributes if its endpoint channel is active.

Definition 4.2 (Boundary normalization). Let

$$W_e(x) = \mu_e(x)([0, \infty)).$$

In the standard unitary three-state branch, $W_e = \sum_i |U_{ei}|^2 = 1$. More general target-compatible source/detector models may have $W_e \neq 1$, but such a change is not a mere mass-spectrum variation; it modifies boundary normalization and must be represented in the target packet or typed as a source/detector update.

4.2 Generalized differential spectrum

Let E be the beta-electron kinetic energy, E_0 the zero-mass endpoint, V_f the excitation energy of molecular final state f , and

$$\epsilon_f(E) = E_0 - V_f - E.$$

Collect the target-fixed smooth factors into

$$\mathcal{K}(E; E_0, V_f) = \frac{G_F^2 |V_{ud}|^2}{2\pi^3} |M_{\text{nuc}}|^2 F(Z, E) p(E + m_e) G(E, E_0 - V_f).$$

The carrier-neutral differential spectrum is

$$R_\beta(E; \mu_e, E_0) = \sum_f P_f \mathcal{K}(E; E_0, V_f) \int_{[0, \epsilon_f(E)^2]} \epsilon_f(E) \sqrt{\epsilon_f(E)^2 - s} d\mu_e(s). \quad (1)$$

The upper integration boundary implements nonnegative neutrino momentum and is the measure-theoretic form of the Heaviside threshold.

Proposition 4.3 (Discrete specialization). *If $\mu_e = \sum_a w_a \delta_{m_a^2}$, then equation (1) becomes*

$$R_\beta(E) = \sum_f P_f \mathcal{K}(E; E_0, V_f) \sum_a w_a \epsilon_f(E) \sqrt{\epsilon_f(E)^2 - m_a^2} \Theta(\epsilon_f(E) - m_a),$$

which is the standard incoherent mass-eigenstate beta spectrum.

Proof. Integration against a finite sum of delta measures evaluates the integrand at each m_a^2 whose support lies below ϵ_f^2 . The support condition is equivalent to the Heaviside factor. $\square$

This formulation isolates the physical spectral object from source labels. Two microscopic realizations with the same complete electron-boundary spectral measure and the same target-fixed source/detector semantics produce the same differential endpoint spectrum before apparatus convolution.

4.3 Standard three-state and effective-mass compression

For the standard three-state branch,

$$d\mu_e(s) = \sum_{i=1}^3 |U_{ei}|^2 \delta(s - m_i^2) ds.$$

When the target response does not resolve the individual thresholds, the collaboration model uses

$$m_\beta^2 = \sum_i |U_{ei}|^2 m_i^2.$$

This is a target-defined compression of the discrete spectral measure. It is not one individual m_i^2 , and it is not the cosmological sum $\sum_i m_i$.

Definition 4.4 (Response-equivalence). Two electron-boundary spectral measures μ and ν are KNM1–5 response-equivalent, written

$$\mu \sim_{K, \text{resp}} \nu,$$

when, after one common lawful choice of target nuisance/calibration coordinates, they induce equal predicted count vectors and equal target-admitted likelihood/acceptance outputs in all seven campaign configurations.

Response equivalence is deliberately stronger than equality of one scalar m_β^2 outside the unresolved branch. Two spectra with the same second moment may produce different resolved kinks or threshold tails.

4.4 Physical support and statistical continuation

The physical measure always satisfies

$$\text{Spec}(\mu_e) \subseteq [0, \infty).$$

The fit parameter $\hat{m}_\beta^2$ is allowed to continue below zero to obtain regular statistical behavior and confidence construction. That continuation changes the fitted response parameter, not the physical support of μ_e .

Lemma 4.5 (No negative physical spectrum). *The reported value $\hat{m}_\beta^2 < 0$ does not imply that any D1 survivor has negative mass-squared spectral support.*

Proof. The target separates the physical source variable μ_e , whose domain is nonnegative by definition and by the square-root threshold in equation (1), from the statistical continuation coordinate used in the likelihood. A negative estimator is a value of the latter, not an element of $\text{Spec}(\mu_e)$. $\square$

4.5 Physical spectral measure after complete endpoint activity

The measure $\mu_e(x)$ is not an independently assignable report label. It is the complete physical electron-boundary spectral measure obtained after every endpoint-potential inherited source contribution has received a typed activity disposition. If the endpoint-active source point has positive squared-kinematic object $K_\beta^{\text{tot}}(x)$, spectral resolution $E_{K_\beta^{\text{tot}}}$, and beta-boundary state $\omega_\beta(x)$, then

$$\mu_e(x)(A) = \langle \omega_\beta(x), E_{K_\beta^{\text{tot}}(x)}(A) \omega_\beta(x) \rangle. \quad (2)$$

Equation (2) is source constitutive whenever such an operator description is present. More general functional/categorical sources carry the corresponding positive measure-valued source object. In either case, the physical spectrum must be derived from the complete source point rather than assigned after the response calculation.

Definition 4.6 (Complete endpoint-activity condition). For a D1 source point x , let $\text{EPot}_\beta(x)$ be the source-constitutive closure of every inherited source object or incidence that can reach the electron beta boundary under the D1 target. Every member is typed as

$$\text{Direct}, \quad \text{Routed}, \quad \text{SourceNull}, \quad \text{TargetUpdate}.$$

Direct contributions enter the total spectral measure. Routed contributions enter only through a source- and target-authorized spectral-preserving route. SourceNull requires a theorem-certified zero boundary incidence. TargetUpdate exits the frozen target. A new D1 completion law may add a further contribution only when it is source coherent and does not overwrite an inherited physical spectrum.

Thus an inherited conventional spectrum

$$\sum_i |U_{ei}|^2 \delta_{m_i^2}$$

remains part of the total beta measure whenever the corresponding mass-bearing path is endpoint active. A route may factor that incidence but cannot replace its support by a freely chosen δ_t . Conversely, a base carrier with no inherited endpoint-kinematic law may admit a genuinely new D1 spectral completion on the same owner. Both cases are generated before target evaluation and are distinguished in the source-faithful completion groupoid of Section 8.

Equality of

$$m_\beta^2 = \int s \, d\mu_e$$

does not imply equality of complete response. Exact unresolved reduction or complete KNM1–5 response equivalence is required wherever the scalar KATRIN likelihood is used.

4.6 Source-semantic completeness of the spectral variable

The PCR-3 source grammar admits local, hidden, composite, solitonic, boundary, environmental, and hybrid carrier architectures with symbolic parameter spaces. PCR-5D1 adds the target-relative beta-source incidence, endpoint-activity disposition, and electron-boundary spectral trace. Any finite occupancy-bearing architecture yields a finite or finitely presented endpoint-potential diagram; functional/categorical source domains yield the corresponding measure-valued family. The source trace retains the microscopic incidence needed for carrier identity, while the endpoint response factors through the complete source-derived μ_e and the declared boundary data.

5 Campaign response, nuisance architecture, and acceptance

5.1 Apparatus response

For campaign c , detector group p , and retarding potential $qU_{c,k}$, let

$$f_{\text{calc}}^{c,p}(E, qU; \eta_c)$$

be the target-fixed probability that an electron emitted with energy E reaches the detector group. Following the KATRIN response framework [1, 3, 5], it is built from the MAC-E transmission, angular source distribution, scattering probabilities, energy-loss densities, synchrotron loss, detector efficiency, field maps, and potential broadening. Schematically,

$$f_{\text{calc}}^{c,p}(E, qU; \eta_c) = \int d\theta d\varepsilon T^{c,p}(E - \varepsilon, \theta, qU; \eta_c) \sin \theta \sum_{s \geq 0} P_s^c(E; \eta_c) f_s^c(\varepsilon; \eta_c). \quad (3)$$

The released inputs instantiate the campaign-dependent parameters η_c and their covariance structure.

The signal rate is

$$S_{c,p,k}(\mu_e, E_0, \eta) = A_c \int R_\beta(E; \mu_e, E_0) f_{\text{calc}}^{c,p}(E, qU_{c,k}; \eta_c) dE, \quad (4)$$

and the predicted released count is

$$\lambda_{c,p,k} = t_{c,k} \epsilon_{c,p,k} \left[S_{c,p,k}(\mu_e, E_0, \eta) + B_{c,p,k}(\eta, \tau_{c,k}^P, \mathcal{T}_c^P) \right]. \quad (5)$$

All campaign data and nuisance coordinates entering this expression are included in the frozen target fingerprint.

5.2 Likelihood object

The primary analysis combines statistically independent campaign/configuration data sets. Nominal-analyzing-plane configurations are treated with the declared normal likelihood approximation, while lower-count shifted-plane data use a Poisson likelihood; systematic uncertainties enter through pull terms and covariance relations [3]. The target-level likelihood may be written

$$-2 \log \mathcal{L}_K(\mu_e, E_0, \eta) = \sum_{c \in \mathcal{G}} \chi_c^2(N_c, \lambda_c) + \sum_{c \in \mathcal{P}} 2 \sum_{p,k} \left[\lambda_{c,p,k} - N_{c,p,k} + N_{c,p,k} \log \frac{N_{c,p,k}}{\lambda_{c,p,k}} \right] + \Pi(\eta), \quad (6)$$

with the usual continuous convention when $N = 0$. The pull/covariance term Π is defined from the released input uncertainties and correlation structures. Equation (6) is a structural target expression; this manuscript does not claim a numerical reimplemention identical to the collaboration analysis code.

5.3 Target acceptance

Definition 5.1 (Frozen standard acceptance). On the collaboration's unresolved standard branch, the target acceptance predicate is

$$\mathcal{A}_K^{\text{UR3}}(m_\beta) = 1 \iff 0 \leq m_\beta \leq m_K, \quad m_K = 0.45 \text{ eV},$$

with 90% confidence semantics imported from the primary analysis. A source-derived one-support measure $\delta_{m_\beta^2}$ realizes this frozen one-parameter response exactly. A multi-support or functional physical spectrum enters the scalar branch only when the source point carries an exact target-authorized unresolved-reduction or complete response-equivalence witness; equality of one moment is insufficient.

For a response-equivalent spectral measure $\mu_e \sim_{K,\text{resp}} \mu_{\text{UR3}}(m_\beta)$, define

$$\mathcal{A}_K(\mu_e) = \mathcal{A}_K^{\text{UR3}}(m_\beta).$$

For response regimes that are not equivalent to the frozen collaboration model, the target returns `ResponseOutOfScope` rather than a physical fail. This rule prevents a standard likelihood from masquerading as an exclusion theorem for a different spectral model.

5.4 Nuisance covariance as one-point structure

The same nuisance point must serve all campaign predictions to the extent required by the published covariance and common-origin relations. Campaign-specific parameters are permitted only where the target declares them. A D1 survivor cannot take the gas density from one source point, the response broadening from another, and the background profile from a third.

Definition 5.2 (Complete campaign response). The complete response of a raw D1 point x is

$$\text{Resp}_K(x) = (\mu_e(x), E_0(x), \eta(x), (\lambda_{c,p,k}(x))_{c,p,k}, \mathcal{L}_K(x), \mathcal{A}_K(x)),$$

together with the provenance and applicability data needed to evaluate every active term.

5.5 Target-complete realization object

First define the empirical KNM1–5 packet

$$T_K^{\text{emp}} = (\mathcal{D}_K, \mathcal{I}_K, \text{Spec}_K, \text{Resp}_K, \mathcal{B}_K, \mathcal{L}_K, \mathcal{C}_K, \mathcal{U}_K, \mathcal{A}_K).$$

Its fields are the measured record, model inputs, differential spectrum, apparatus response, background, likelihood and statistical semantics, calibration and reference data, licensed empirical continuations, and acceptance predicate. The complete realization target is

$$\hat{T}_K^{\text{BB}} = (T_{\text{osc}}, T_K^{\text{emp}}, \mathbb{S}_\beta^{\text{BB}}, \text{SOE-D1-1}, \mathcal{C}_\beta^{\text{BB}}, \mathcal{A}_\beta^{\text{BB}}, \mathcal{W}_\beta, \mathcal{U}_\beta, \Xi_\beta^{\text{BB}}, \mathcal{N}_\beta^{\text{BB}}, Q_\beta^{\text{BB}}).$$

Thus the target contains not only the empirical record but also the versioned source ontology, comparison class, candidate and survivor groupoids, witness and continuation structures, standing packet, standing-core groupoid, and quotient on which the D1 theorems are evaluated. Throughout the paper, the abbreviation T_K refers to this complete object; T_K^{emp} refers only to the empirical packet.

Theorem 5.3 ([D1-01] KATRIN enriched-target completeness). *The frozen object $\hat{T}_K^{\text{BB}}$ is target-complete*

for the KNM1–5 direct-kinematic comparison class consisting of the collaboration unresolved branch and every spectral measure response-equivalent to it. The object contains every load-bearing public record, response, calibration, nuisance, covariance, background, final-state, likelihood, applicability, and update coordinate used by that target.

Proof. The data component follows from [theorem 3.1](#). The spectrum, response, and statistical factors are bound to the cited primary and methods sources; the final-state and calibration refinements are bound to the corresponding version-locked papers. The acceptance predicate is fixed before carrier evaluation. Every raw source branch has one of three target-level dispositions: it lies in the unresolved standard branch, is response-equivalent to that branch, or changes the response/source/detector class and is typed as `ResponseOutOfScope` or `TargetUpdate`. Cosmological, Majorana/LNV, heavy-state-search, and symmetry coordinates are absent by construction. Hence the comparison class and every standing-bearing coordinate are exhausted. $\square$

Direct falsifier

A load-bearing campaign field, covariance block, response term, background class, final-state input, calibration relation, or statistical convention used by the frozen KNM1–5 analysis is absent or reinterpreted; or a nonstandard response is called excluded without a target-defined likelihood.

6 Response-regime partition and target-relative scope

A source-complete lift domain must include endpoint architectures that are not described by the frozen one-parameter KATRIN likelihood. Target completeness does not require that likelihood to answer a different spectral question; it requires every proposal to receive a typed disposition.

6.1 The exact frozen scalar branch

Definition 6.1 (Unresolved standard branch UR3). A point lies in UR3 when its complete differential spectrum, seven-campaign apparatus response, and likelihood object are exactly those of the version-locked KATRIN one-parameter model at some physical value $m_\beta^2 = t \geq 0$. A source-derived one-support physical measure

$$\hat{\mu}_e = \delta_t$$

is an exact realization of this branch. It is not a universal replacement for inherited multi-support endpoint activity.

A conventional multi-support spectrum

$$\mu_e = \sum_i w_i \delta_{m_i^2}$$

may be compressed phenomenologically to an effective $m_\beta^2 = \sum_i w_i m_i^2$ in an unresolved approximation. In the present exact target calculus it belongs to UR3 only when the source carries a target-authorized exact unresolved-reduction witness; otherwise it must pass the complete response-equivalence test below. Equality of the second moment alone is not exact response equality.

Definition 6.2 (Response-equivalent branch `RespEq`). A spectral measure lies in `RespEq` when its complete differential spectrum after lawful target reduction, all seven campaign predicted counts, and the target-admitted likelihood/acceptance object are isomorphic to those of a UR3 point after lawful nuisance/calibration descent.

Definition 6.3 (Resolvable-kink branch **Kink**). A point lies in **Kink** when one or more electron-coupled thresholds produce a target-relative endpoint kink not represented by the frozen one-parameter likelihood.

Definition 6.4 (Functional spectral branch **Functional**). A point lies in **Functional** when its target-active endpoint response requires a continuous or functional spectral measure not response-equivalent to the frozen scalar branch.

The final primary regime consists of altered source, detector, weak-operator, analysis-window, or confidence-construction semantics. These are target updates rather than alternative spectra inside the frozen target.

6.2 Disposition theorem

Proposition 6.5 (Response-regime disposition). *Every source-generated D1 proposal receives exactly one primary disposition*

UR3, RespEq, Kink, Functional, TargetUpdate,

*followed, where applicable, by the complete response/acceptance status. The **Kink** and **Functional** branches are **ResponseOutOfScope** unless an exact isomorphism to a frozen response object or a target-qualified likelihood is supplied.*

Proof. The electron-boundary trace either realizes the exact one-parameter target model, induces an equal complete response, produces a finite target-relative threshold structure beyond that model, requires a non-discrete response, or changes the beta-decay source/detector/operator/statistical semantics. These cases are disjoint by construction and exhaustive for the declared finite-or-functional comparison class. $\square$

6.3 Why out-of-scope is not elimination

A resolvable heavy-state kink, a broad continuum, or an altered weak interaction can be physically viable while lying outside the target used to derive the quoted 0.45 eV limit. The correct implication is

response model changes $\implies$ successor target required,

not

standard fit is unavailable $\implies$ carrier impossible.

This distinction is load-bearing for D1-02, D1-05, and D1-11.

6.4 Decoupled heavy states

A heavy state with squared mass s larger than every available endpoint energy satisfies

$$\epsilon_f(E)^2 < s$$

throughout the analysis window and contributes zero to the standard spectral kernel. If its only D1 effect is through that inaccessible endpoint channel, it is response-inert. It may nevertheless remain a standing-distinct base carrier because PCR-5B matching or continuation witnesses see its threshold. This separation prevents endpoint invisibility from collapsing base SCIC identity.

7 Absolute-scale geometry and response-admissible strata

7.1 Hyperprocess absolute-scale branch

For every framed PCR-5B source survivor H , the declared enriched-source constructor BB acts on the upstream source hypergraph before outcome evaluation. The Source-Authorized Process-Compilation Theorem proves that the resulting hyperprocess presentation is source coherent, retains irreducible higher incidence, has current-source deletion behavior identical to H , and has the same complete oscillation standing core after evaluation. The D1 metric law then gives

$$K_{\beta,t} = t\mathbf{1}, \quad \mu_{\beta,t} = \delta_t, \quad t \in J_{\beta}^{\text{BB}} := [0, m_K^2).$$

Theorem 8.22 identifies this branch with the exact frozen one-parameter KNM1–5 model at $m_{\beta}^2 = t$. Hence every base residue contains a continuum branch.

7.2 Intrinsic conventional common-offset families

Fix an intrinsic conventional multi-support branch with weights $w_i \geq 0$, $\sum_i w_i = 1$, and relative squared-mass offsets δ_i with $\min_i \delta_i = 0$. When its source chart admits a lawful common-offset family,

$$m_i^2(t) = t + \delta_i, \quad m_{\beta}^2(t) = t + \bar{\delta}_e, \quad \bar{\delta}_e := \sum_i w_i \delta_i.$$

Theorem 7.1 (Conditional conventional unresolved scale interval). *Assume an exact target-authorized unresolved-reduction witness or complete response isomorphism to the frozen scalar KATRIN model. Then the physical 90% common-offset locus is*

$$I_{\beta}^{90}(w, \delta) = [0, m_K^2 - \bar{\delta}_e]$$

when $\bar{\delta}_e \leq m_K^2$, and is empty otherwise.

Proof. Under the stated response witness the frozen scalar acceptance applies to $m_{\beta}^2(t)$. Combining $t \geq 0$ with $t + \bar{\delta}_e \leq m_K^2$ gives the interval. Equality of moments without a response witness is insufficient. $\square$

7.3 Source-fixed, source-open, and class-level scale

The D1 source atlas distinguishes:

1. source-fixed intrinsic presentations evaluated on their exact physical spectra;
2. source-open intrinsic provenance families with lawful common-offset or other scale moduli;
3. response-out-of-scope or target-update presentations; and
4. the explicit enriched-source hyperprocess-envelope branch, obtained from an upstream B15 source hypergraph by the explicitly declared SOE-D1-1 constructor and carrying no inherited absolute beta spectral object.

For an intrinsic source point z , the exact accepted locus $J_{\beta}(z)$ may be empty, singleton, finite, interval-valued, functional, categorical, or mixed. The class residue nevertheless contains J_{β}^{BB} through the distinct compiled provenance point $\text{BB}(H)$.

7.4 Zero floor, positive floor, and nonstandard spectra

Every source-faithful intrinsic completion is classified by its actual physical support:

- zero-lightest boundary: $\inf \text{supp } \mu_e = 0$;
- positive-floor stratum: $\inf \text{supp } \mu_e > 0$;
- parameterized-scale stratum;
- resolvable-kink stratum;
- functional/continuous stratum;
- mixed stratum.

These enrich, rather than replace, the compiled one-support component.

Proposition 7.2 (Absolute-support separation). *Let two accepted D1 residue points over one base class have complete physical measures μ and ν . If their absolute supports or complete KNM1–5 response objects differ after lawful nuisance/calibration descent, they are distinct enriched SCIC classes. A difference only in source chart, basis, nuisance coordinate, or response-inert provenance is not a split.*

Proof. Absolute spectral support and complete endpoint response are admitted D1 standing consequences. PCR-5C standing descent identifies points only when every retained enriched consequence agrees; provenance-only differences are quotiented by the lawful D1 groupoid. $\square$

The classwise nonempty-residue theorem proves that every base class survives and contains continuum many hyperprocess scale points. High-offset intrinsic presentations may fail simultaneously. This coexistence is expected because SCIC is target relative and source provenance is not carrier identity.

Direct beta-decay standing does not select Dirac/Majorana character, field inventory, ordering, or a unique ultraviolet mechanism [8, 13].

8 The enriched D1 source category and the nonempty residue

The direct beta-endpoint target activates a new metric use of the retained electron boundary. The proof of universal survival must respect two independent orderings:

source construction precedes outcome evaluation, constraint exhaustion precedes residue claims.

PCR-3 and the PCR-5B B15 handoff generate the source domain. The D1 evaluator then imposes the frozen KATRIN target together with source, boundary, continuation, matching, failure, and restriction constraints. AASC fixes the admissibility and identity consequences of that evaluation and closes null, skin-authority, outside-governance, and same-act-repair pseudo-routes. It does not manufacture a source point. The nonempty residue proved below is supplied by an independently defined, source-authorized hyperprocess construction inside the explicitly declared D1 enriched-source ontology.

8.1 Source-faithful endpoint activity

Fix a PCR-5B family/skeleton chart (F, σ) and a source-coherent base point $z \in Z_{F, \sigma}$. Let

$$\text{EPot}_\beta(z) \subseteq \text{Trace}_{\text{src}}(z)$$

be the source-constitutive closure of every node, law, mass map, matching arrow, hidden projection, constituent/profile map, boundary incidence, environmental coupling, or higher relation that can acquire an electron-boundary beta incidence under the frozen D1 target.

Definition 8.1 (Endpoint-activity disposition). Every object $a \in \text{EPot}_\beta(z)$ receives exactly one status

Direct, Routed, SourceNull, TargetUpdate.

Direct means that the inherited source object supplies an active beta-boundary spectral contribution. *Routed* means that the same contribution is transported through an explicitly declared beta incidence without altering its physical spectral content. *SourceNull* requires a source theorem proving zero beta-boundary weight. *TargetUpdate* means that the proposed incidence changes the frozen weak/source interaction, detector semantics, response operator, or another target primitive. No endpoint-potential source object may be omitted.

For each Direct or Routed object let

$$\mu_{\beta,a}^{\text{src}}(z) \in \text{Meas}_+([0, \infty))$$

denote its source-derived electron-boundary contribution.

Definition 8.2 (Spectral-preserving route). A route

$$\varrho_{\beta,a} : a \longrightarrow \text{Own}_e(z)$$

is D1-lawful only when it preserves the physical contribution of a up to a declared source- and target-authorized spectral isomorphism. Thus

$$(\varrho_{\beta,a})_* \mu_{\beta,a}^{\text{src}}(z) \cong \mu_{\beta,a}^{\text{src}}(z),$$

and the electron-boundary weight is preserved. A route that replaces $\sum_i w_i \delta_{m_i^2}$ by $W \delta_t$ without a constitutive equality is not a route in the frozen target; it is a source or target update.

Let $D(z)$ and $R(z)$ be the Direct and Routed subsets. For any genuinely new D1 completion datum η admitted by the enriched source grammar, define

$$\mu_\beta^{\text{tot}}(z, \eta) = \sum_{a \in D(z)} \mu_{\beta,a}^{\text{src}}(z) + \sum_{a \in R(z)} (\varrho_{\beta,a})_* \mu_{\beta,a}^{\text{src}}(z) + \mu_\beta^{\text{new}}(\eta). \quad (7)$$

Theorem 8.3 (Source-faithful endpoint-activity theorem). *Every lawful D1 extension of a PCR-5B source point determines a total endpoint-activity partition and a complete physical measure of the form (7). Conversely, a source-coherent completion datum together with a total partition, lawful routes, the frozen beta boundary, and the D1 response equations defines a raw D1 source point. Forgetting D1-only structure returns the declared base reduct.*

Proof. PCR-3 generates the complete source trace before survival, and B15 supplies the solved source chart together with its source-equational groupoid. D1 activates precisely the beta-boundary incidences. Direct and spectrally preserving Routed contributions remain in the physical measure, SourceNull terms vanish only by theorem, and TargetUpdate terms leave the fixed target. The resulting positive measure is therefore the complete electron-boundary spectral object entering the carrier-neutral response. D1-only symbols are removed by base restriction while inherited source equations remain fixed. $\square$

Corollary 8.4 (Active-singlet spectral preservation). *For a conventional active-singlet presentation with*

$$\kappa_N = Y_\nu M_R^{-1} Y_\nu^\top, \quad \mu_e^{3N} = \sum_i |U_{ei}|^2 \delta_{m_i^2},$$

any Direct or Routed D1 completion retains the multi-support contribution up to a source-authorized spectral isomorphism. It may not be replaced by a one-support measure without an independent constitutive equality.

8.2 The complete class-indexed pre-evaluation universe

Let $q \in Q_{\text{osc}}^{\text{SCIC}}$ and let $\text{Prov}_{\text{osc}}(q)$ be its complete framed B15 source-provenance groupoid. For each framed source presentation (z, α) , let $\mathcal{C}_\beta(z)$ be the groupoid of every source-coherent D1 completion admitted by the version-locked D1 enriched grammar before KATRIN evaluation. It includes intrinsic source-derived spectra, lawful absolute-scale continuations, functional or continuous spectra, hidden/composite/profile completions, finite hybrids, and the hyperprocess-compiled branch constructed below.

Definition 8.5 (Candidate-complete D1 lift universe). The pre-evaluation D1 universe over q is

$$\widetilde{\mathcal{L}}_\beta(q) := \int_{(z, \alpha) \in \text{Prov}_{\text{osc}}(q)} \widetilde{\mathcal{C}}_\beta(z). \quad (8)$$

Its objects retain the framed base provenance, all new source data, endpoint activity, physical spectral measure, response data, and every source/target status required to decide survival.

Theorem 8.6 (Candidate-complete lift-universe theorem). *Equation (8) contains every lawful same-target D1 carrier realization with base restriction in q , up to enriched source equivalence. It is formed independently of the KATRIN pass/fail result and is invariant under deletion or alteration of every downstream outcome.*

Proof. PCR-5C requires the complete raw lift domain to be generated before evaluation. PCR-3 source completeness covers every finite active source-incidence architecture and every source-defined symbolic parameter object in the declared comparison class; B15 supplies every base provenance presentation. The D1 grammar, including the explicitly declared extension SOE-D1-1, adds exactly the endpoint, matching, response, nuisance, calibration, and enriched-source coordinates permitted by the frozen target and declared comparison class. Taking the Grothendieck construction over the entire framed provenance groupoid therefore includes every valid base reduct and every lawful enriched completion without choosing a preferred model or survivor. $\square$

8.3 The simultaneous D1 constraint bundle

For $y \in \widetilde{\mathcal{L}}_\beta(q)$ define

$$C_\beta(y) = (C_{\text{src}}, C_{\text{base}}, C_{\text{bdy}}, C_{\text{resp}}, C_K, C_{\text{cont}}, C_{\text{match}}, C_{\text{fail}}, C_{\text{SCIC}})(y). \quad (9)$$

The nine entries respectively test source-constitutive coherence, complete base-packet retention, charged-current endpoint ownership, response validity, KNM1–5 likelihood/acceptance, lawful continuation/activity, matching/gluing/common-point coherence, fail-closed/no-retuning semantics, and restriction to the declared base SCIC class.

Definition 8.7 (D1 residue). The standing-positive residue over q is

$$\mathcal{R}_\beta(q) := \{y \in \widetilde{\mathcal{L}}_\beta(q) : \mathcal{C}_\beta(y) = \mathbf{1}\}. \quad (10)$$

After lawful enriched presentation descent,

$$\mathcal{R}_\beta(q) \simeq \mathcal{L}_\beta(q).$$

Definition 8.8 (Exact complement partition). The complement of (10) is partitioned into saturated typed subgroupoids

$$\mathcal{F}_\beta(q) = F_{\text{phys}} \sqcup F_{\text{src}} \sqcup F_{\text{target}} \sqcup F_{\text{null}} \sqcup F_{\text{repair}} \sqcup F_{\text{skin}} \sqcup F_{\text{undef}}.$$

The cells record respectively genuine fixed-target physical incompatibility, source-constitutive failure, target/source update, null or reconstruction-vacuous pseudo-lifts, same-act repair, skin/provenance promoted to authority, and unresolved/undefined response or applicability.

Theorem 8.9 (AASC constraint-partition theorem). *The cells F_{null} , F_{repair} , and F_{skin} cannot contain an admissible same-target standing lift. An outside-but-governing selector, AMetric default, hidden same-act reservoir, or provenance veto lies in one of those cells or in F_{target} . The cells F_{phys} , F_{src} , F_{target} , and F_{undef} remain genuine possible negative dispositions and are not removed by AASC closure.*

Proof. The null-regime closure theorem excludes total vacuity from the positive side of an already evaluated same-type admissibility problem; it does not infer an occupant from an empty fragment. Realized-persistence non-vacuity applies only to a standing-bearing continuation already in force. Bivalence, ATS role exhaustion, no hidden selector, and no same-act repair therefore close the three pseudo-lift cells and the corresponding scope-changing variants. Source incoherence, physical mismatch, undefinedness, and target update are legitimate failures under PCR-4/PCR-5C and remain in the complement. $\square$

8.4 The D1 source comparison class

The universal operation needed by the D1 witness is not silently attributed to PCR-3 saturation. The frozen PCR-3 algebra contains law/process objects, derived nodes, finite hybrids, and irreducible higher incidence, but it does not state a universal constructor sending an arbitrary coherent carrier hypergraph to an endpoint-neutral hyperprocess presentation. PCR-5D1 therefore declares one explicit *enriched-target* source-ontology extension. The frozen oscillation source theory, packet, and quotient are not changed.

Definition 8.10 (D1 hyperprocess-envelope source extension SOE-D1-1). Let $\mathbb{S}_{\text{car}}(T_{\text{osc}})$ be the frozen PCR-3 source theory and let $\mathbb{S}_\beta(T_K)$ denote the independently typed raw D1 source theory. Define

$$\mathbb{S}_\beta^{\text{BB}}(T_K) := \mathbb{S}_\beta(T_K)[\text{HypProc}, \text{BB}, \text{Del}^{\text{BB}}, U_\beta^{\text{BB}}]/\mathcal{E}_{\text{BB}}. \quad (11)$$

The extension adds exactly:

1. a hyperprocess source sort **HypProc** on an already occupancy-bearing electron-boundary carrier owner;
2. a constructor **BB** from finite activity-reduced source-coherent PCR-3 carrier hypergraphs to enriched hyperprocess objects;
3. current-source intervention functors Del_S^{BB} for every source-authorized constitutively closed deletion set S ;

4. a base-reduct functor U_β^{BB} satisfying

$$U_\beta^{\text{BB}}(\text{BB}(H)) = H;$$

5. the source-only equations $\mathcal{E}_{\text{BB}}$ of typing, source coherence, preservation of primitive arity and higher coherence, intervention naturality, boundary/ownership covariance, matching and gluing covariance, endpoint opacity, and base-reduct functoriality.

No PCR-4 result, standing packet, survivor label, failure record, KATRIN datum, or desired endpoint value occurs in the signature or in $\mathcal{E}_{\text{BB}}$. In particular, packet equality is not a formation axiom. It is proved downstream in Theorem 8.17. PCR-5D1 makes no claim that BB was already among the frozen PCR-3 operations.

Let $G_{\text{car}}^{\text{coh},\beta,\text{BB}}(T_K)$ denote the finite source-coherent models of (11). For such a model Y , write

$$\Phi_4^{\text{ret}}(Y) := \Phi_4(U_\beta^{\text{BB}}(Y)), \quad \Xi_{\text{osc}}^{\text{ret}}(Y) := \Xi_{\text{osc}}(U_\beta^{\text{BB}}(Y)).$$

These are downstream retained-target evaluations; they are not constructor inputs.

Exact comparison-class declaration

The universal result below is relative to the explicit D1 enriched-source extension SOE-D1-1. The base source ontology $\mathbb{S}_{\text{car}}(T_{\text{osc}})$, the PCR-5B standing packet, and $Q_{\text{osc}}^{\text{SCIC}}$ remain frozen. The extension is removed by the declared base-reduct functor and therefore does not add, delete, split, or merge an oscillation-level standing class.

Definition 8.11 (Source-coherent hyperprocess family). Write

$$\Theta_{\text{hproc}}^{\text{src}} := \{\text{BB}(H) : H \in G_{\text{car}}^{\text{coh}}(T_{\text{osc}})\} / \simeq_{\text{src}}$$

for the source-coherent participating family generated in the extended source theory by the constructor BB, its current-source intervention arrows, and the lawful source-equational arrows of $\mathbb{S}_\beta^{\text{BB}}(T_K)$. The family is generated only after SOE-D1-1 has been declared; it is not attributed to the frozen PCR-3 operation list.

8.5 The source-authorized hyperprocess envelope

Let H be a finite pointwise activity-reduced PCR-3 source-coherent carrier hypergraph,

$$H = (V_H, E_H, \lambda_H, \omega_H, \Theta_H, a_H),$$

where E_H includes ordinary and irreducible higher-arity incidence and ω_H contains every source-constitutive and higher-coherence cell.

Definition 8.12 (Colored hyperprocess envelope). The colored hyperprocess $\mathbb{P}(H)$ is the finite colored multicategorical presentation generated by:

1. one color/port for every current source kind, carrier owner, boundary, matching, continuation, and target-attachment type of H ;
2. one primitive operation $[e]$ for every $e \in E_H$, with the same arity, typing, support, and orientation as

e ;

3. the parameter object Θ_H , transported without numerical or semantic alteration; and
4. exactly the relations, associators, module/junction identities, derivation cells, overlap diagrams, and higher-coherence cells ω_H .

No irreducible hyperedge is decomposed into pairwise paths. The envelope is finite because H is finite at the frozen PCR-3 scope.

Definition 8.13 (Source-authorized hyperprocess compilation). Let c_H be the existing electron-boundary carrier owner fixed by the source attachment. Define

$$\text{BB}(H) := (c_H, \mathbb{P}(H), \Sigma_H, \mathcal{B}_H, \mathcal{G}_H, \Omega_H, \mathcal{M}_H, \mathcal{X}_H, a_H; H).$$

Here Σ_H is the current-source intervention action induced by deleting colored ports and every operation whose constitutive support meets the deleted set, followed by source-constitutive reclosure. The remaining entries are the inherited boundary/ownership, representation-charge-anomaly, vacuum/deformation, oriented matching, extension/gluing, and target-attachment data. The final occurrence of H is the exact base-reduct object used by U_β^{BB} ; it is not a second D1-active microscopic sector and contributes no beta spectral support.

Lemma 8.14 (Outcome blindness of the extension). *Formation of $\text{BB}(H)$ is invariant under deletion or alteration of every PCR-4/PCR-5/D1 outcome object. It depends only on the source hypergraph, source parameters, constitutive equations, activity support, and frozen source attachment.*

Proof. Every constructor argument occurs in the PCR-3 source trace or in the version-locked enriched-source signature SOE-D1-1. That signature has no slot for a survivor label, standing core, packet value, failure result, or KATRIN output. Hence the source object exists, when H exists, before downstream evaluation. $\square$

Theorem 8.15 (Source coherence and irreducible-incidence preservation). *For every $H \in G_{\text{car}}^{\text{coh}}(T_{\text{osc}})$,*

$$\text{BB}(H) \in G_{\text{car}}^{\text{coh}, \beta, \text{BB}}(T_K).$$

The constructor preserves every primitive node-kind port, every source parameter needed by the retained oscillation process, and every ordinary or irreducible higher-arity incidence and coherence relation. In particular, a fusion, common-environment, inflow, collective, or genuinely many-body hyperedge remains one primitive hyperprocess operation of the same arity.

Proof. The raw hyperprocess is generated by the new source sort and constructor of SOE-D1-1. Its constitutive equations are the source equations ω_H already satisfied by H , transported along the arity-preserving assignment $e \mapsto [e]$. Hence all source identities and higher-coherence cells commute. No factorization of a non-factorizing edge is used. Boundary, owner, matching, vacuum, gluing, and target-attachment data are transported by the typed source maps declared in $\mathcal{E}_{\text{BB}}$. Therefore the result is a finite model of the explicitly extended enriched-source theory. $\square$

8.6 Current-source deletion and retained-target evaluation

For a constitutively closed source-kind deletion set S , let $\text{Del}_S(H)$ be the frozen PCR-5A/PCR-5B no-repair intervention on the current source presentation. It fixes all retained parameters, vacuum,

matching, boundary, gluing, and continuation data.

Theorem 8.16 (Current-source deletion naturality). *For every source-authorized deletion set S ,*

$$\text{Del}_S^{\text{BB}}(\text{BB}(H)) \cong \text{BB}(\text{Del}_S(H)) \quad (12)$$

as current enriched-source hyperprocess objects, with the appropriate source-null or failure value when the deleted source no longer realizes the retained role. The intervention acts on the ports and primitive operations of the compiled current source, not on a provenance label.

Proof. Deleting S removes exactly the colors belonging to S , every operation whose constitutive support requires one of those colors, and every dependent coherence cell. Source-constitutive reclosure retains exactly the operations generated by the no-repair retained subhypergraph. Since the envelope construction is functorial on source restriction and copies each irreducible edge before quotienting, compilation commutes with deletion. No parameter is reoptimized and no replacement route is inserted. $\square$

Theorem 8.17 (Source-Authorized Process-Compilation Theorem). *For every $H \in G_{\text{car}}^{\text{coh}}(T_{\text{osc}})$, the D1 enriched-source point $\text{BB}(H)$ is source coherent and satisfies*

$$\Phi_4^{\text{ret}}(\text{BB}(H)) = \Phi_4(H), \quad \Xi_{\text{osc}}^{\text{ret}}(\text{BB}(H)) = \Xi_{\text{osc}}(H). \quad (13)$$

Moreover,

$$\Delta_{\text{fr}}^{\text{del,ret}}(\text{BB}(H)) = \Delta_{\text{fr}}^{\text{del}}(H),$$

where the left-hand side is evaluated by interventions lawful on the compiled current source. Every irreducible higher incidence remains represented by its primitive hyperprocess operation.

Proof. Source coherence and higher-arity preservation are Theorem 8.15. The base-reduct equation in SOE-D1-1 gives

$$U_{\beta}^{\text{BB}}(\text{BB}(H)) = H.$$

The retained-target evaluator therefore receives exactly the original PCR-4 event/process, boundary, common-cone, continuation, matching, representation/anomaly, vacuum, extension, and failure inputs. This proves the first equality in (13); PCR-5A/PCR-5B then give equality of all seven standing channels. The deletion channel is additionally computed from Theorem 8.16, so it is a current-source result rather than copied history. The evaluator equalities are thus conclusions of the declared source extension and its reduct, not formation arguments. $\square$

Theorem 8.18 (Source-level endpoint neutrality). *Before a D1 metric law is attached,*

$$\text{EPot}_{\beta}^{\text{abs}}(\text{BB}(H)) = \emptyset.$$

The compiled enriched-source point carries retained oscillation-process structure, intervention ports, and the frozen ledgers, but no object of the absolute beta spectral sort, no beta-boundary spectral attachment, and no D1 metric operator. Intrinsic mass-bearing presentations of H remain separate source-provenance objects in the candidate-complete D1 universe; they are not overwritten or reclassified as the hyperprocess point.

Proof. Endpoint opacity is a source-typing clause of HypProc: the process parameters are not typed as beta mass-support objects, and the base-reduct object H is not D1-active. Since no AbsSpec_{β} object or beta spectral attachment occurs in the active hyperprocess trace, the absolute endpoint-potential

extractor returns the empty object. This is a property of the new enriched-source architecture, not a claim that the original microscopic presentation lacked intrinsic kinematics. $\square$

Theorem 8.19 (Conservative base-reduct theorem). *The explicit source extension leaves the frozen oscillation source universe and standing quotient unchanged. For every H ,*

$$U_{\beta}^{\text{BB}}(\text{BB}(H)) = H, \quad [U_{\beta}^{\text{BB}}(\text{BB}(H))]_{\text{SCIC}} = [H]_{\text{SCIC}}.$$

Thus no old base class is removed or merged and no new oscillation-level standing class is introduced. The hyperprocess object exists only in the enriched D1 candidate universe and restricts to the frozen base source point.

Proof. The first equality is one of the version-locked source equations of SOE-D1-1. The second follows from the complete packet equality in Theorem 8.17. Because base restriction lands in the frozen PCR-3 source universe, the base quotient is not replaced by an extended quotient. $\square$

Corollary 8.20 (Compiled D1 provenance over every base core). *For every $q \in Q_{\text{osc}}^{\text{SCIC}}$ and every B15 framed source point (H, α) over q , the D1 enriched provenance fibre contains $\text{BB}(H)$ and its base reduct is framed over q . No standing core is used to form the source point; the frame is used only after source formation.*

8.7 Exact D1 completion of the hyperprocess branch

For $t \geq 0$, let $\mathbb{M}_{\beta,t}^{\text{1sp}}$ be the one-support D1 metric law on the same electron-boundary carrier owner,

$$K_{\beta,t} = t\mathbf{1}, \quad \mu_{\beta,t} = \delta_t.$$

It adds no primitive carrier node and no second terminal role.

Definition 8.21 (Hyperprocess D1 completion). For a source point H , define

$$\mathfrak{E}_{\beta,t}^{\text{BB}}(H) := \text{BB}(H) \amalg_{\text{Own}_e(H)} \mathbb{M}_{\beta,t}^{\text{1sp}},$$

with the frozen tritium source, electron boundary, KNM1–5 response, nuisance, likelihood, and acceptance objects attached to the compiled carrier owner.

Theorem 8.22 (Exact hyperprocess KNM1–5 response). *For every source-coherent target-participating H and every*

$$t \in J_{\beta}^{\text{BB}} := [0, m_K^2), \quad m_K = 0.45 \text{ eV},$$

the complete physical endpoint measure of $\mathfrak{E}_{\beta,t}^{\text{BB}}(H)$ is δ_t . Its differential spectrum, seven-campaign count prediction, nuisance/covariance object, likelihood, and acceptance are exactly the frozen one-parameter KNM1–5 objects at $m_{\beta}^2 = t$. If H is a base survivor, the completed point is a D1 survivor.

Proof. Theorem 8.18 proves that the active hyperprocess source has no inherited absolute beta spectral object. The one-support law is therefore the complete D1 endpoint measure of this source presentation. Substitution of δ_t into the carrier-neutral beta functional gives the exact one-parameter differential model, and every frozen final-state, apparatus, background, covariance, nuisance, likelihood, and acceptance map is retained. The version-locked upper-limit target admits every $t < m_K^2$. Theorem 8.17 supplies the retained oscillation standing packet. $\square$

Lemma 8.23 (Exact base neutrality). *For every base survivor H and $t \in J_\beta^{\text{BB}}$,*

$$r_\beta^{\text{src}}(\mathfrak{E}_{\beta,t}^{\text{BB}}(H)) = H, \quad \Xi_{\text{osc}}(r_\beta \mathfrak{E}_{\beta,t}^{\text{BB}}(H)) = \Xi_{\text{osc}}(H).$$

Proof. Forgetting D1 removes the one-support metric law, response objects, and hyperprocess wrapper; the declared base-reduct functor returns H . The packet equality is Theorem 8.17. No source parameter is retuned after evaluation. $\square$

8.8 Constraint-induced residue and universal splitting

Theorem 8.24 (D1 constraint-induced nonempty residue theorem). *For every $q \in Q_{\text{osc}}^{\text{SCIC}}$,*

$$\mathcal{R}_\beta(q) \neq \emptyset.$$

For every chosen B15 framed source point (H_q, α_q) over q and every $t \in J_\beta^{\text{BB}}$,

$$[\mathfrak{E}_{\beta,t}^{\text{BB}}(H_q)] \in \mathcal{R}_\beta(q).$$

Proof. The enriched source point is generated by the explicitly declared source ontology and is source coherent by Theorems 8.15 and 8.17. The exact response theorem proves KNM1–5 acceptance, and base neutrality proves restriction to q . Hence all nine D1 constraint channels hold on one source-coherent point. AASC enters only afterward to close null, skin, hidden-selector, and same-act-repair pseudo-dispositions. $\square$

Corollary 8.25 (Marked-class calibration). *Each of $q_W, q_{3N}, q_{\text{red}}, q_{\text{syn}}$ has a D1 survivor for every $t \in J_\beta^{\text{BB}}$. Current-source deletion on the hyperprocess presentations recovers the frozen pairs $(0, 1), (1, 0), (1, 1), (0, 0)$ by Theorem 8.16. The original intrinsic presentations remain separate provenance objects with their source-derived endpoint spectra.*

Theorem 8.26 (Universal direct-kinematic choice-section and fibre theorem). *Choose once and for all one B15 framed source presentation $\chi(q) = (H_q, \alpha_q)$ for each $q \in Q_{\text{osc}}^{\text{SCIC}}$. For every $t \in J_\beta^{\text{BB}}$ define*

$$s_t^\chi(q) := [\mathfrak{E}_{\beta,t}^{\text{BB}}(H_q)].$$

Then

$$\rho_\beta \circ s_t^\chi = \text{id}_{Q_{\text{osc}}^{\text{SCIC}}}.$$

For each fixed q , the map

$$J_\beta^{\text{BB}} \longrightarrow \pi_0 \mathcal{L}_\beta(q), \quad t \longmapsto s_t^\chi(q)$$

is injective. Consequently

$$\text{im } \rho_\beta = Q_{\text{osc}}^{\text{SCIC}} \tag{14}$$

and

$$|\pi_0 \mathcal{L}_\beta(q)| \geq \mathfrak{c} \quad \forall q \in Q_{\text{osc}}^{\text{SCIC}}. \tag{15}$$

Different choices of framed provenance may define different enriched sections; no representative-independence claim is required or made.

Proof. The right-inverse identity is Lemma 8.23. If $t \neq t'$, the enriched absolute-support channel contains the unequal physical measures δ_t and $\delta_{t'}$. This is D1 standing rather than provenance, so SCIC forbids

their identification. Since J_β^{BB} is a nondegenerate real interval, it has cardinality $\mathfrak{c}$. $\square$

Remark 8.27 (Exact role of AASC). AASC neither creates SOE-D1-1 nor proves its source objects inhabited. The source extension and hyperprocess constructor are explicitly declared and source coherence is proved before KATRIN evaluation. AASC constrains the evaluated residue and excludes null, skin, hidden-governance, outside-but-governing, and same-act-repair pseudo-dispositions.

8.9 Intrinsic branches and negative control

For an arbitrary source point z , define

$$\text{Comp}_\beta^K(z) = \{\eta : \mathcal{A}_K(\text{Resp}_K(\mu_\beta^{\text{tot}}(z, \eta))) = 1\}.$$

Intrinsic discrete, kinked, functional, continuous, hidden, composite, solitonic, environmental, boundary, topological, and mixed spectra populate additional fibre components or typed complement strata.

Theorem 8.28 (High-offset presentation exclusion). *There exist source presentations whose source-derived physical endpoint response fails the frozen KATRIN acceptance branch. Such failure removes that completion but does not eliminate its base SCIC class because the explicit hyperprocess-envelope completion supplies another pre-generated enriched presentation in the same standing fibre. No failed source is repaired after evaluation.*

Remark 8.29 (Constraint-native conclusion). The universal result is relative to the declared enriched-source extension and is not inferred from null-fibre incoherence. It follows from a source-only hyperprocess constructor, current-source deletion naturality, exact base reduct, endpoint neutrality, a fixed D1 response, and complete constraint evaluation. The source-ontology change is visible and version locked rather than attributed silently to frozen PCR-3 saturation.

9 Lift domains over the oscillation standing quotient

9.1 Raw endpoint extensions over a base class

For $q \in Q_{\text{osc}}^{\text{SCIC}}$, the complete framed provenance groupoid is $\text{Prov}_{\text{osc}}(q)$. The pre-evaluation universe is the Grothendieck construction

$$\widetilde{\mathcal{Z}}_\beta(q) = \int_{(z, \alpha) \in \text{Prov}_{\text{osc}}(q)} \widetilde{\mathcal{C}}_\beta(z)$$

of Definition 8.5. It contains every intrinsic endpoint completion and every SOE-D1-1 hyperprocess completion before KATRIN survival is read.

At a concrete base core $n \in \mathcal{N}(T_{\text{osc}})[q]$, the equivalent source-level pullback is

$$\mathcal{C}_\beta^{\text{raw}}(n) = \mathcal{C}_{j_\beta}^{\text{red}}(\mathbb{S}_\beta^{\text{BB}}(T_K)) \times_{\mathcal{C}(T_{\text{osc}})}^h \widetilde{\text{Prov}}_0(n). \quad (16)$$

The declared base-reduct functor lands in the frozen PCR-3 source groupoid by Theorem 8.19. The pullback therefore ranges over all valid base reducts without replacing or enlarging the base carrier space.

9.2 A universal hyperprocess subfibre

For every framed B15 source survivor (H, ϕ) , Theorem 8.17 constructs the source point $\text{BB}(H)$ inside the explicitly declared source extension. Its D1 metric completions form

$$\mathcal{C}_\beta^{\text{BB}}(H) = \{\text{BB}(H)\} \times J_\beta^{\text{BB}}, \quad J_\beta^{\text{BB}} = [0, m_K^2]. \quad (17)$$

Theorem 8.22 places this object inside the standing residue, and Equation (13) proves that its base standing core is the core of H .

Lemma 9.1 (No-side-carrier discipline). *The D1 metric law acts on the electron-boundary carrier owner of $\text{BB}(H)$ and creates no second primitive carrier or terminal role. An independently occupancy-bearing beta-only node is a separate HYB completion and remains separately represented.*

Lemma 9.2 (Endpoint-anchor base neutrality). *For every intrinsic completion (z, η) with valid base reduct and every hyperprocess completion,*

$$\Xi_{\text{osc}}(r_\beta(z, \eta)) = \Xi_{\text{osc}}(z), \quad \text{Core}_5(r_\beta \mathfrak{C}_{\beta, t}^{\text{BB}}(H)) \cong \text{Core}_5(H).$$

Endpoint-only standing can split a D1 fibre but cannot alter the frozen base packet.

9.3 Raw restriction statuses

The PCR-5C raw restriction classifier returns

ValidBasePoint, BaseNull, UpstreamReopening, TargetUpdate, Undefined.

Only ValidBasePoint objects enter a fibre over q . A source/detector interaction change, spectral overwrite, or altered weak operator is TargetUpdate. SOE-D1-1 is not hidden in this classifier: its source operation is declared and proved conservative before the lift calculation.

Theorem 9.3 ([D1-02] Direct-kinematic lift-domain completeness). *For every $q \in Q_{\text{osc}}^{\text{SCIC}}$:*

1. $\widetilde{\mathcal{L}}_\beta(q)$ is source complete relative to the explicitly frozen D1 comparison class, including SOE-D1-1;
2. the simultaneous constraint bundle partitions it exactly into the residue and the typed complement of Definition 8.8;
3. the subfibre (17) is generated from an upstream PCR-3/B15 source hypergraph by the declared source-only constructor BB , not from a standing core and not from AASC null exclusion; and
4. every point of that subfibre is an accepted D1 source point.

Hence the raw base fibre contains a nonempty accepted component before intrinsic endpoint branches are quotient-assembled.

Proof. Source completeness is Theorem 8.6; the total partition is Theorem 8.9; source formation, coherence, deletion naturality, and packet recovery are Theorems 8.15, 8.16, and 8.17; exact acceptance is Theorem 8.22. AASC is used only to type and close inadmissible complement cells. $\square$

9.4 Response-domain completeness versus likelihood completeness

A generated Kink or Functional proposal remains present even when the frozen scalar likelihood is undefined on it. It receives the exact response-domain or successor-target status. The hyperprocess branch is evaluated by the exact one-support response and does not replace any intrinsic component.

Direct falsifier

A valid endpoint architecture is absent from the raw universe; SOE-D1-1 is used without being declared; BB consults a standing core, interface result, survivor label, or KATRIN outcome; an irreducible higher hyperedge is path-factorized; current-source deletion fails Equation (12); one of the seven packet channels is not recovered; the enriched-source extension fails to reduce to the frozen base source point or changes a base SCIC class; the compiled source contains an inherited absolute beta spectral object; the D1 metric law is attached to a second carrier; or one compiled point fails the exact KNM1–5 target.

10 One-point KNM1–5 survival

10.1 Complete enriched witness packet

A raw D1 source point carries the frozen PCR-5B packet and applicability record, a complete electron-boundary spectral measure, all seven campaign predictions, endpoint/normalization/background/final-state/calibration variables, covariance relations, target likelihood/acceptance, and source/response/update statuses.

Theorem 10.1 ([D1-03] One-point KNM1–5 survivor theorem). *A raw D1 point is a survivor if and only if one common source point:*

1. *restricts to a PCR-5B survivor in the declared base class;*
2. *realizes the complete frozen oscillation packet;*
3. *carries the complete source-faithful endpoint measure of its D1 completion;*
4. *produces the frozen KNM1–5 spectrum and campaign count predictions;*
5. *satisfies the nuisance, covariance, calibration, likelihood, and acceptance relations;*
6. *carries no active source, response, applicability, failure, null/skin, repair, or update defect.*

For every framed B15 source survivor H , the hyperprocess-envelope completion $\mathfrak{E}_{\beta,t}^{\text{BB}}(H)$ formed in SOE-D1-1 satisfies these conditions for every $t \in J_{\beta}^{\text{BB}}$.

Proof. Necessity is the fixed-target one-point condition inherited from PCR-4/PCR-5C. Sufficiency follows because the listed conditions are exactly the nine components of $\mathcal{C}_{\beta}$. The extended-source point is source coherent, higher-incidence faithful, and packet faithful by Theorems 8.15, 8.16, and 8.17; it has exact response by Theorem 8.22 and exact base restriction by Lemma 8.23. $\square$

10.2 Residue and complement are not interchangeable

The complete D1 calculation contains two logically different objects:

1. the standing residue $\mathcal{R}_{\beta}(q)$, whose objects satisfy all simultaneous constraints;

2. the exact complement partition $\mathcal{F}_\beta(q)$, which retains genuine physical/source failures, undefinedness, updates, null/skin pseudo-lifts, and same-act repair attempts.

AASC closes some complement cells as impossible positive-side statuses. It does not turn a point in a genuine source or physical failure cell into a survivor.

10.3 Hyperprocess-envelope and intrinsic survivor families

The full survivor atlas contains:

1. the *hyperprocess-envelope branch* in the explicit enriched-target source extension, which proves nonempty residue and continuum splitting over every base class;
2. *intrinsic source-faithful branches*, which retain all source-derived discrete, kinked, functional, continuous, composite, solitonic, hidden, environmental, boundary, topological, and mixed endpoint spectra and evaluate them by the frozen response or an exact complement status.

High-offset intrinsic presentations may fail. Their failure does not contradict a hyperprocess-envelope survivor in the same base SCIC class because the intrinsic source and $\text{BB}(H)$ are distinct pre-evaluation provenance presentations in the declared enriched comparison class.

10.4 No splicing, no packet lookup, no silent ontology change, and no null inference

A base source, D1 completion, endpoint measure, and KATRIN response may not be assembled from unrelated candidates. The constructor BB starts from one source presentation, retains every constituent and irreducible source relation in one colored hyperprocess envelope, and attaches the D1 metric law to the same owner. Intrinsic completions satisfy the same one-point rule.

The extension SOE-D1-1 and its source equations are fixed before PCR-4/PCR-5/D1 evaluation. A standing core is used only to range over its already existing B15 provenance fibre after the source point has been formed. No source output coefficient is set equal to a solved packet coordinate, and no claim is made that the black-box constructor was already in frozen PCR-3.

Nor may one infer survival by arguing that an empty residue would be null. The nonempty residue is proved by the explicit source-extension model, its conservative-base theorem, and its D1 completion before AASC null closure is applied.

Direct falsifier

The source extension is undeclared or falsely attributed to frozen PCR-3 saturation; the constructor or its parameter domain changes after reading an interface value, standing packet, failure set, survivor label, or KATRIN result; an irreducible higher incidence is flattened; deletion is evaluated on provenance history rather than the compiled current source; $\text{BB}(H)$ fails one source equation or one packet channel; the extension changes the frozen base quotient; the hyperprocess envelope already contains an absolute beta spectral object; the endpoint measure differs from the measure fed into the response; or a failed point is retuned after evaluation and relabelled as the same candidate.

11 Chartwise lift atlas

The D1 atlas is indexed by every PCR-5B family/skeleton chart and its complete symbolic parameter domain; it is not a finite list of named models. The source extension SOE-D1-1 supplies an additional hyperprocess provenance presentation over each source point, while the intrinsic chart remains present as

a separate component.

For each (F, σ) let

$$\tilde{Z}_{\beta;F,\sigma}$$

be the complete raw D1 lift object and

$$Z_{\beta;F,\sigma}^K$$

its simultaneous standing residue. The raw object includes all intrinsic source-faithful completions, every target-lawful D1 source extension, all response and nuisance data, and the compiled branch

$$\text{BB}(Z_{F,\sigma}) \times J_{\beta}^{\text{BB}}.$$

Theorem 11.1 (Executed chartwise D1 atlas). *For every family/skeleton chart:*

1. *the raw D1 lift domain is candidate complete relative to the declared source ontology;*
2. *the nine-part D1 constraint bundle and seven-cell complement partition are total;*
3. *the hyperprocess subobject is source coherent, higher-incidence faithful, current-source deletion natural, endpoint neutral, and nonempty over every source point;*
4. *the complete standing residue is the union/codescent of the accepted hyperprocess and intrinsic components under the lawful enriched groupoid;*
5. *every connected base core represented by the chart survives; and*
6. *each base fibre contains at least continuum many enriched SCIC components.*

The result is symbolic and functorial over the complete parameter object; no finite sampling is used.

Proof. Items 1–2 are the candidate-universe and constraint-partition theorems. Item 3 follows from the source-ontology extension, Source-Authorized Process-Compilation Theorem, deletion naturality, endpoint neutrality, and exact response theorem. Item 4 is lawful groupoid descent. Items 5–6 follow from the choice-section and absolute-support injection. The quantifiers range over every PCR-5B skeleton and source parameter point represented by the chart. $\square$

11.1 Family dispatch

The six family chart types are

$$\text{LAW}, \quad \text{LOCAL}, \quad \text{HID}, \quad \text{COMP}, \quad \text{SOL}, \quad \text{HYB}.$$

They are atlas charts rather than carrier identities. A concrete standing core can occur in more than one chart, and the overlap calculus identifies duplicate enriched presentations only when the complete D1 standing core agrees.

For every family, the chart data are

$$(\tilde{Z}_{\beta;F,\sigma}, \mathcal{C}_{\beta;F,\sigma}, \mathcal{F}_{\beta;F,\sigma}, Z_{\beta;F,\sigma}^K, \Gamma_{\beta;F,\sigma}^K, p_{\beta;F,\sigma}^K, \Delta_{\beta;F,\sigma}^{\text{rel}}).$$

These objects retain the source-extension status, current-source intervention action, higher-incidence structure, intrinsic endpoint activity, and every response or failure disposition.

11.2 Base-moduli partition

Because every base point receives an accepted real interval, each chart has

$$\mathcal{M}_{F,\sigma}^\emptyset = \mathcal{M}_{F,\sigma}^1 = \mathcal{M}_{F,\sigma}^{\text{finite}} = \mathcal{M}_{F,\sigma}^{\text{countable}} = \mathcal{M}_{F,\sigma}^{\text{NotReady}} = \emptyset,$$

while

$$\mathcal{M}_{F,\sigma}^{\text{ParameterizedMixed}} = \mathcal{M}_{F,\sigma}.$$

This is a theorem over the exact chart moduli, not an extrapolation from the four marked examples.

Failure conditions for atlas completeness

A valid B15 skeleton or parameter stratum is omitted; the new source operation is not explicitly registered; higher incidence is flattened; deletion is copied from provenance instead of evaluated on $\text{BB}(H)$; one chart lacks a compiled accepted interval; or a finite table replaces the symbolic chart quantifier.

12 Restriction, no merger, and choice-dependent sections

The KATRIN target induces the exact restriction functor $R_{\beta,\text{BB}}$ and quotient map $\rho_{\beta,\text{BB}}$. Every enriched SCIC class lies over exactly one base class, and distinct base classes never merge. The enriched-source extension adds D1 provenance presentations but its declared base-reduct functor lands exactly in the frozen PCR-3 source universe; the base quotient is never replaced.

Theorem 12.1 ([D1-04] Restriction and no-merger theorem). *There is a well-defined functor*

$$R_{\beta,\text{BB}} : \mathcal{N}(\hat{T}_K^{\text{BB}}) \longrightarrow \mathcal{N}(T_{\text{osc}})$$

and

$$\rho_{\beta,\text{BB}} = \pi_0(R_{\beta,\text{BB}}) : Q_K \longrightarrow Q_{\text{osc}}^{\text{SCIC}}.$$

If $q \neq q'$, no enriched SCIC class restricts to both. For any choice function

$$\chi : q \longmapsto (H_q, \alpha_q)$$

selecting one B15 framed source presentation over each base class and every $t \in J_\beta^{\text{BB}}$, the map

$$s_t^\chi(q) := [\mathfrak{C}_{\beta,t}^{\text{BB}}(H_q)]$$

satisfies

$$\rho_{\beta,\text{BB}} \circ s_t^\chi = \text{id}_{Q_{\text{osc}}^{\text{SCIC}}}.$$

No claim is made that two different choices χ produce the same enriched section.

Proof. Representative-independent restriction and no merger are inherited from PCR-5C witness retention. Theorem 8.19 proves that every hyperprocess completion has the exact frozen PCR-3 base reduct. For the chosen frame, Theorem 8.22 gives an accepted D1 point and Lemma 8.23 gives the right-inverse identity. Distinct base classes retain their PCR-5B separators, so no enriched equivalence can bridge them. Different source presentations of one base class may become D1-standing-distinct; only a choice-dependent section is required for surjectivity. $\square$

12.1 Restriction of intrinsic and compiled components

For every intrinsic D1 completion (z, η) with valid base reduct,

$$r_{\beta}^{\text{src}}(z, \eta) = z.$$

For every compiled completion,

$$r_{\beta}^{\text{src}}(\mathfrak{C}_{\beta,t}^{\text{BB}}(H)) = H.$$

The exact base-reduct theorem therefore returns the frozen source point and its standing class.

The operation BB is not a source-equivalence claim. It creates a new source provenance presentation whose complete base packet is proved equal only after evaluation. The original intrinsic source point remains available and can produce additional D1 lifts or failures.

12.2 Frozen deletion on the current compiled source

The source-kind intervention functor of SOE-D1-1 is part of the current hyperprocess source. For every deletion set S ,

$$\text{Del}_S^{\text{BB}} \text{BB}(H) \cong \text{BB}(\text{Del}_S H).$$

Therefore the four marked classes retain

$$q_W : (0, 1), \quad q_{3N} : (1, 0), \quad q_{\text{red}} : (1, 1), \quad q_{\text{syn}} : (0, 0).$$

The intervention does not consult provenance history and does not retune after deletion.

Restriction falsifier

A compiled D1 point restricts to a packet-distinct base class; two base-separated classes merge; deletion is evaluated from the original provenance rather than from the compiled current source; the source extension fails to return the exact frozen base source point or changes its base SCIC class; or the proof requires different B15 presentations of one base class to yield one identical enriched section.

13 Lift-fibre exhaustion and classwise residue

For $q \in Q_{\text{osc}}^{\text{SCIC}}$, the exact D1 lift groupoid is

$$\mathcal{L}_{\beta, \text{BB}}(q) = \mathcal{N}(\hat{T}_K^{\text{BB}}) \times_{\mathcal{N}(T_{\text{osc}})}^h \mathcal{N}(T_{\text{osc}})[q].$$

The base leg is the frozen PCR-5B standing-core groupoid because the declared D1 base-reduct functor returns the original PCR-3 source point. Its source presentation is the descended residue $\mathcal{R}_{\beta}(q)$ of Definition 8.7.

The standing-positive residue contains, for any selected B15 frame (H_q, α_q) ,

$$\{[\mathfrak{C}_{\beta,t}^{\text{BB}}(H_q)] : t \in J_{\beta}^{\text{BB}}\}. \quad (18)$$

Intrinsic accepted components are assembled alongside this interval and retain their complete source-derived spectra, automorphisms, response objects, and failure frontiers.

Theorem 13.1 ([D1-05] Exact fibre exhaustion). *For every $q \in Q_{\text{osc}}^{\text{SCIC}}$,*

$$\pi_0 \mathcal{L}_{\beta, \text{BB}}(q) \cong \rho_{\beta, \text{BB}}^{-1}(q) \cong \pi_0 \mathcal{R}_{\beta}(q).$$

The fibre is nonempty and contains an injection

$$J_{\beta}^{\text{BB}} \hookrightarrow \pi_0 \mathcal{L}_{\beta, \text{BB}}(q).$$

Hence every base fibre has coarse type `ParameterizedMixed` and cardinal lower bound at least $\mathfrak{c}$. The global enriched quotient is

$$Q_K \cong \bigsqcup_{q \in Q_{\text{osc}}^{\text{SCIC}}} \pi_0 \mathcal{L}_{\beta, \text{BB}}(q).$$

No upper cardinal independent of the frozen comparison class is claimed.

Proof. The first identity is the PCR-5C homotopy-fibre theorem; the second is the exact residue descent. Theorem 8.24 supplies the compiled point over every selected frame and every $t \in J_{\beta}^{\text{BB}}$. Lemma 8.23 identifies its base class. Unequal t give unequal admitted endpoint-support measures, hence distinct enriched SCIC components. Intrinsic components can add further points but cannot remove the compiled interval. Every enriched class has one base restriction, yielding the disjoint global assembly. $\square$

13.1 Provenance-level decomposition

For a framed base source point (z, α) let

$$\mathcal{R}_{\beta}(z) = \mathcal{R}_{\beta}^{\text{int}}(z) \sqcup \mathcal{R}_{\beta}^{\text{BB}}(z)$$

denote its intrinsic and hyperprocess-derived accepted components. Codescent over the complete provenance groupoid gives

$$\mathcal{R}_{\beta}(q) \simeq \underset{(z, \alpha) \in \text{Prov}_{\text{osc}}(q)}{\text{codesc}} \mathcal{R}_{\beta}(z).$$

A failed intrinsic presentation removes only its own summand. The base class survives because the selected provenance point admits the separately generated SOE-D1-1 completion. This conclusion depends on the explicit enriched-source extension and its exact base-reduct theorem, not on AASC null exclusion.

13.2 Intrinsic scale loci

For a source presentation with a lawful intrinsic absolute-scale continuation, let $J_{\beta}^{\text{int}}(z)$ be its exact response-admissible locus. Every such component embeds into the intrinsic D1 subfibre after lawful source quotienting. These components can be empty, singleton, finite, continuous, functional, categorical, or mixed. They coexist with the universal hyperprocess component and are not quotiented into it unless their complete enriched standing cores agree.

Fibre falsifier

A base class has no selected B15 source frame; SOE-D1-1 is not part of the declared candidate universe; the compiled source fails source coherence, deletion naturality, endpoint neutrality, or KNM1–5 acceptance; unequal t are standing-equivalent; or an intrinsic failure is promoted to class elimination despite the accepted compiled component.

14 Calibration, nuisance, estimator, and parameter descent

14.1 Four parameter roles

The KNM1–5 target contains many coordinates, but not every coordinate is a standing-carrier coordinate. For a candidate and fixed target, each parameter variation has one primary role:

1. **Target anchor.** It changes the data, unit, detector/source semantics, response model, analysis window, or confidence construction. The result is a successor target.
2. **Lawful experimental redescription.** It changes chart, gauge, unit representation, or an equivalent calibrated presentation while preserving the complete response and failure behavior.
3. **Standing response parameter.** It changes an admitted endpoint-response, matching, boundary, continuation, or failure consequence.
4. **Nuisance/provenance parameter.** It is integrated, profiled, constrained, or recorded in the target uncertainty architecture but does not refine the carrier once the complete standing response is fixed.

The classification is determined by target use rather than by the word “systematic” or “nuisance.” A calibration quantity may be standing at the experimental target if changing it changes the realized response, yet it need not be part of microscopic carrier identity once response-equivalent presentations are quotiented.

14.2 Lawful response equivalence

Definition 14.1 (D1 response-equivalence relation). Two enriched survivor presentations x, y are response-equivalent when there exists a target-lawful presentation arrow carrying:

- the complete frozen base standing packet;
- the electron-boundary spectral object up to endpoint-response equivalence;
- every campaign prediction;
- the joint covariance and calibration object;
- source-to-endpoint matching and boundary ownership;
- acceptance, continuation, and failure behavior;

from x to y .

Equality of a scalar estimator is not enough. Conversely, different internal nuisance coordinates may define one response-equivalence class.

14.3 Parameter-descent theorem

Theorem 14.2 ([D1-06] KATRIN parameter-descent theorem). *Experimental nuisance, calibration, chart, estimator, and presentation coordinates refine enriched SCIC identity if and only if they alter an admitted standing endpoint-response consequence after lawful continuation. Otherwise they are quotiented by D1 response equivalence or retained solely as target uncertainty/provenance data.*

Proof. SCIC fixes carrier identity by complete standing continuation signatures. A parameter variation that changes target definedness or semantics is an anchor and lies outside the same target. With the target fixed, a variation that changes a campaign response, endpoint spectral consequence, boundary ownership, matching, acceptance, or same-target failure changes the enriched standing signature. If it changes none of these, retaining it as a separate carrier identity would require an unwitnessed second classifier. It may remain in the experimental uncertainty or source-provenance record without refining the terminal carrier quotient. $\square$

14.4 Disposition of the released input sectors

Table 2 gives the default role of the major public input sectors. The word “default” matters: a candidate-specific cross-relation can make a normally nuisance-level coordinate standing, but that promotion must be proved by an admitted target witness.

Table 2: Default D1 parameter dispositions.

Coordinate family	Default level	Standing criterion
Retarding voltages and units	Target/calibration anchor	Change of physical scan coordinate or lawful unit transport
Live times and efficiencies	Target response/nuisance	Change of expected campaign counts after lawful calibration
Electromagnetic fields	Target response/calibration	Change of transmission or resolution visible to the endpoint record
Gas density and energy loss	Target response/nuisance	Change of scattering response or failure status
Isotopologue concentrations and final states	Spectrum/response	Change of the endpoint spectral shape
Rear-wall and background terms	Target background/nuisance	Change of target likelihood or response failure
Source-potential and drift parameters	Response/calibration	Change of energy shift/broadening after allowed transport
Negative $\hat{m}_\beta^2$ estimator	Inference coordinate	Never negative physical support; standing only through the frozen interval construction
Source field inventory or mechanism label	Provenance by default	Standing only if a target witness distinguishes the complete endpoint response

14.5 Nulls, signed inputs, and finite precision

The audit preserves three transmission-broadening uncertainty entries as explicit nulls. A null means the uncertainty was not supplied in that field; it does not mean zero uncertainty. Signed potential-drift broadening-squared fit inputs remain signed in the response model and are not reinterpreted as negative physical spectral mass. The finite asymmetry of the gas-density covariance remains part of source provenance, while mathematical covariance use must state whether it uses the released matrix, its symmetrized form, or another declared continuation.

Proposition 14.3 (No silent normalization). *Replacing null uncertainties by zero, clipping signed fitted inputs, or silently symmetrizing a released covariance changes the target representation unless the transformation is declared and proved response-equivalent.*

Proof. Each operation changes the numerical input object. It is lawful only if included in C_K with proof

that the complete response and inference standing are preserved. Otherwise it changes the target or analysis continuation. $\square$

15 Source-composition refinement

15.1 No-repair deletion at the enriched target

Let x be a D1 survivor with finite active source-block set $K_\beta(x)$. For $S \subseteq K_\beta(x)$, retain exactly S while holding fixed every surviving source parameter, the electron boundary, vacuum branch, matching orientation, gluing, response regime, target record, and calibration/nuisance coordinate of the parent point. No replacement source, spectral overwrite, retuning, or re-gluing is permitted.

Definition 15.1 (D1 source-sufficiency function). Define

$$N_{\beta,x} : 2^{K_\beta(x)} \rightarrow \{0, 1, \perp\}$$

by $N_{\beta,x}(S) = 1$ when the fixed deletion point remains a complete D1 survivor, 0 when it fails, and $\perp$ when the deletion is not constitutively well typed. Let

$$\text{MinSuff}_\beta(x) = \{S : N_{\beta,x}(S) = 1 \text{ and no proper subset is sufficient}\}.$$

15.2 S/R/P/Y taxonomy

The exact source organization is:

- S (single-route) when there is one minimal sufficient route and it is not the unique full multi-source set;
- R (redundant) when there are multiple minimal routes and at least one singleton route;
- P (partially synergistic) when there are multiple minimal routes and every minimal route is multi-source;
- Y (synergistic) when $|K_\beta(x)| > 1$ and the full active set is the unique minimal sufficient route.

Theorem 15.2 ([D1-07] Direct-kinematic source-composition theorem). *For every finite-source D1 survivor, $\text{MinSuff}_\beta(x)$ determines exactly one of S, R, P, Y . The classification is invariant under lawful source redescription, uses no retuning, and refines enriched SCIC identity only through target-admitted standing consequences.*

Proof. The minimal sufficient subsets form a finite nonempty antichain. The four cases above are disjoint and exhaustive by the number and cardinalities of its members. Lawful source isomorphisms preserve the deletion incidence and therefore the antichain. SCIC forbids the source-organization label from acting as an independent classifier when all D1 standing consequences agree. $\square$

15.3 Endpoint-active blocks are source derived

The D1 source set contains every inherited endpoint-active block from $\text{EPot}_\beta(x)$ together with any genuinely new D1 completion blocks. A Routed block may only factor an inherited endpoint incidence through a spectral-preserving route in the sense of Definition 8.2. It cannot replace an inherited mass spectrum by a more convenient one, and it cannot repair a deleted oscillation route.

For a deletion trial, the complete endpoint measure is recomputed from the retained source blocks:

$$\mu_{\beta,S}^{\text{tot}} = \sum_{a \in D_S} \mu_{\beta,a}^{\text{src}} + \sum_{a \in R_S} (\varrho_{\beta,a})_* \mu_{\beta,a}^{\text{src}} + \mu_{\beta,S}^{\text{new}},$$

then passed through the frozen KNM1–5 evaluator. This makes source composition a semantic calculation rather than a label on a preselected response.

Proposition 15.3 (Base-deletion conservativity). *After all D1-only blocks are removed, the frozen PCR-5B deletion channel is exactly recovered. In particular, whenever an accepted D1 completion lies over the marked redundant or synergistic calibration sectors, their LAW/new-local deletion pairs remain (1, 1) and (0, 0) respectively. A D1 endpoint block is never counted as a replacement for a deleted base oscillation route.*

Proof. D1-04 recovers the exact base source point and packet. The no-repair deletion test is then evaluated on that frozen point, with all parent parameters held fixed. A D1-only block is target-inert after restriction and therefore cannot enter the base sufficiency calculation. $\square$

15.4 Class-level versus presentation-level composition

Source-composition profiles are carried in the D1 provenance record. Different source presentations of one base SCIC class may have different endpoint-activity antichains while restricting to the same frozen packet. They refine the enriched carrier only if the D1 target witnesses a resulting response, matching, continuation, boundary, or failure difference. A failed source presentation cannot be promoted into base-class elimination unless every provenance presentation and lawful D1 completion of that class has been exhausted as required by Equation (23) and Theorem 19.1.

16 Exact base image and selection strength

Let

$$\mathcal{J}_{\beta}^{\text{exec}} := \text{essIm}(R_{\beta,\text{BB}}) \subseteq \mathcal{N}(T_{\text{osc}})$$

be the full subgroupoid of base standing cores admitting at least one accepted D1 residue.

Theorem 16.1 ([D1-08] Exact full-base-image theorem). *For the frozen KNM1–5 target and declared SOE-D1-1 source class,*

$$\boxed{\mathcal{J}_{\beta}^{\text{exec}} = \mathcal{N}(T_{\text{osc}}), \quad \text{im } \rho_{\beta,\text{BB}} = Q_{\text{osc}}^{\text{SCIC}}.} \quad (19)$$

Equivalently,

$$\boxed{\text{codesc}_{[k] \in \Delta^{\text{op}}} \mathcal{P}_{\beta,k}^K \simeq \mathcal{N}(T_{\text{osc}}).} \quad (20)$$

The direct-kinematic target therefore eliminates no base SCIC class and selects no proper base subspace.

Proof. For every concrete base core choose one B15 framed source presentation. The nonempty-residue theorem supplies its accepted hyperprocess completion and exact base frame, so every base object lies in the essential image. The reverse inclusion is tautological. The exact base-reduct theorem places every hyperprocess completion over its frozen base core, while the six residue charts and all overlap equalizers glue through exact core isomorphisms. $\square$

16.1 Selection hierarchy

PCR-5C distinguishes presentation exclusion, base-class elimination, base selection, and full enriched singleton selection. PCR-5D1 proves that presentation exclusion occurs but base-class elimination does not. Since $|Q_{\text{osc}}^{\text{SCIC}}| \geq \mathfrak{c}$,

$$|\text{im } \rho_{\beta, \text{BB}}| = |Q_{\text{osc}}^{\text{SCIC}}| \geq \mathfrak{c},$$

so neither base selection nor enriched singleton selection occurs.

16.2 Why full image is still a target-relative result

Full image does not mean that every intrinsic source presentation survives or that KATRIN adds no standing. Intrinsic points may be excluded, response-out-of-scope, undefined, or target-updating. The theorem says that after the complete framed provenance fibre, the explicitly declared enriched-target source extension, and all D1 completions are exhausted, every frozen base class has a standing-positive residue. The absolute-support coordinate then refines every class into continuum many enriched classes.

17 Universal splitting and absolute-scale refinement

A base class $q \in Q_{\text{osc}}^{\text{SCIC}}$ is D1-split when $|\pi_0 \mathcal{L}_{\beta, \text{BB}}(q)| > 1$. A split is standing-bearing only when two accepted completions differ in an admitted D1 consequence after lawful nuisance/calibration descent.

Theorem 17.1 ([D1-09] Universal direct-kinematic split theorem). *For every $q \in Q_{\text{osc}}^{\text{SCIC}}$, two accepted residue points define distinct enriched SCIC classes exactly when their complete D1 standing signatures differ. For any fixed B15 provenance choice over q , the hyperprocess completions at unequal parameters $t, t' \in J_{\beta}^{\text{BB}}$ have unequal physical endpoint-support measures*

$$\delta_t \neq \delta_{t'}.$$

Therefore

$$\boxed{|\pi_0 \mathcal{L}_{\beta, \text{BB}}(q)| \geq \mathfrak{c} \quad \forall q \in Q_{\text{osc}}^{\text{SCIC}}.} \quad (21)$$

Every PCR-5B base class is split by direct absolute-support standing.

Proof. The first assertion is enriched SCIC. The choice-section theorem supplies the accepted interval in every fibre. Absolute endpoint support is an admitted enriched standing evaluator, not a source label. No target-lawful response, nuisance, or source redescription identifies distinct Dirac measures δ_t and $\delta_{t'}$. Since J_{β}^{BB} has cardinality $\mathfrak{c}$, the result follows. $\square$

17.1 Compiled and intrinsic components

The hyperprocess residue supplies one continuum split in every fibre. Intrinsic source-derived branches may add discrete, continuous, functional, categorical, kinked, or mixed components. They remain separate unless their complete enriched standing signatures agree. In particular, an intrinsic multi-support spectrum is not collapsed into the one-support compiled branch by family name or equality of an effective second moment.

17.2 Zero and positive floor

The compiled branch contains the standing-distinct loci $t = 0$ and $t > 0$. For intrinsic branches, the analogous conditions

$$\inf \text{supp } \mu_e = 0, \quad \inf \text{supp } \mu_e > 0$$

are standing distinct only if both survive the complete target and remain unequal after lawful response descent. An ordering label alone is not a separator.

No ordering, Dirac/Majorana character, field count, ultraviolet familiarity, naturalness, or simplicity criterion enters D1-09.

18 Intersections of enriched carrier charts

For enriched charts $a = (F, \sigma)$ and $b = (G, \tau)$, define

$$\mathcal{O}_{\beta;a,b} = \Gamma_{\beta;a}^K \times_{\mathcal{N}(\widehat{T}_K^{\text{BB}})}^h \Gamma_{\beta;b}^K. \quad (22)$$

An object consists of two accepted D1 source presentations together with an isomorphism of their complete enriched standing cores. Equality includes the frozen seven-channel base packet and, at D1 resolution, the complete endpoint spectral object, source-to-endpoint matching, response regime, campaign prediction/likelihood, parameter-descent class, and failure and update data.

Theorem 18.1 ([D1-10] Executed enriched-overlap theorem). *All fifteen unordered family-pair overlap bands are the homotopy fibre products (22) over the concrete enriched standing-core groupoid. The forty-two distinct-family higher sectors and every repeated-label finite tuple are the corresponding iterated homotopy fibre products. Their complement strata retain exact physical/source failure, response-out-of-scope, null/skin/repair, undefined, source-update, target-update, and theorem-certified N/A data. Codescent reconstructs $\mathcal{N}(\widehat{T}_K^{\text{BB}})$.*

Proof. B15 supplies the complete base charts and provenance frames. The D1 compiler forms the raw objects, applies the simultaneous constraint bundle, and retains only standing-positive objects. Compiled overlap requires a common base core, a common absolute parameter, and equality of the complete hyperprocess source-extension data. Intrinsic overlap requires equality of the complete source-derived enriched standing signature. Associativity of framed homotopy pullback gives the higher nerve, and PCR-5C codescent reconstructs the enriched core groupoid. $\square$

18.1 Compiled and intrinsic overlap sectors

The hyperprocess subfibre contributes one common- t overlap sector wherever two base charts represent the same concrete standing core and the selected source presentations are connected through the exact base overlap. Intrinsic sectors retain their complete physical spectra and response objects. A multi-support intrinsic spectrum and a one-support compiled spectrum coincide only if their complete enriched standing signatures are target-lawfully isomorphic.

Current-source deletion equality in an overlap is checked by the compiled intervention functor, not by a copied historical deletion label. A failure of deletion naturality, higher-incidence preservation, or exact base-reduct preservation is therefore an overlap failure rather than a permitted provenance difference.

18.2 No finite-registry inference

The fifteen and forty-two counts enumerate family-label incidence types, not scientific candidates. Each incidence type quantifies over every generated skeleton pair or tuple and carries its exact symbolic or categorical equalizer. Semantic completeness comes from B15 coverage, the D1 candidate-universe theorem, the declared enriched-target hyperprocess extension, residue closure, and the framed overlap construction.

19 Failure, undefinedness, and structural closure

19.1 Typed failure cells

Every raw D1 point receives the endpoint-activity disposition of Definition 8.1, the PCR-5C raw-restriction status, and the complete nine-component D1 constraint status. The failure object distinguishes at least:

- **SourceOntologyExtensionUndeclared**: the construction uses a source sort, operation, or constitutive equation not present in frozen PCR-3 and not declared by an explicit extension;
- **FrozenPCR3Attribution**: the new hyperprocess-envelope operation is falsely attributed to PCR-3 saturation rather than to SOE-D1-1;
- **OutcomeBackflow**: a standing core, packet value, interface coordinate, pass/fail status, survivor label, expected occupant, or KATRIN result enters source formation;
- **HyperprocessCoherenceFailure**: the compiled envelope violates a source-constitutive equation, attachment condition, activity relation, representation/charge constraint, vacuum condition, matching relation, or gluing equation;
- **HigherIncidenceFlattening**: an irreducible fusion, inflow, shared constraint, common-environment incidence, or higher-arity source relation is replaced by a merely pairwise/pathwise surrogate;
- **CurrentSourceDeletionFailure**: the frozen kindwise deletion is evaluated only on provenance history rather than as a lawful intervention on the compiled current source;
- **DeletionNaturalityFailure**: $\text{Del}_S^{\text{BB}}(\text{BB}(H)) \not\cong \text{BB}(\text{Del}_S H)$ for an admitted frozen deletion S ;
- **ConservativeQuotientFailure**: the source extension adds, deletes, merges, or splits oscillation standing cores rather than presenting the same frozen quotient;
- **EndpointNeutralityFailure**: the compiled envelope already contains a beta-active absolute spectral object before the D1 metric law is attached;
- **EvaluatorFactorizationFailure**: $\Phi_4^{\text{ret}}(\text{BB}(H)) \neq \Phi_4(H)$ or one of $T, R, \Delta_{\text{fr}}^{\text{del}}, C, M, B, F$ changes after complete evaluation;
- **PostCensusCandidateInsertion**: a source family or parameter point is inserted only after a downstream outcome is known;
- **RecordMismatch**: the predicted count/likelihood object fails the frozen KNM1–5 target;
- **ResponseOutOfScope**: the frozen likelihood is not defined on the response object;
- **NullPositiveSide**: a null or reconstruction-vacuous pseudo-lift is assigned standing;
- **SkinAuthority**: presentation or provenance is promoted to a gate or carrier selector;

- **SameActRepair**: a failed source is retuned or replaced while treated as the same evaluated act;
- **TargetUpdate**: the completion changes weak/source/detector or other target semantics;
- ordinary source, matching, continuation, boundary, undefined, applicability, source-update, and upstream-reopening failures inherited from the base atlas.

19.2 What AASC closes and what it does not

AASC closes **NullPositiveSide**, **SkinAuthority**, hidden-selector, AMetric-default, outside-but-governing, and **SameActRepair** as positive-side survivor statuses. It does not supply the hyperprocess constructor, prove source coherence by itself, or infer an occupant from the inadmissibility of total vacuity. Candidate formation is performed by the source grammar enlarged explicitly by SOE-D1-1; the D1 and AASC constraints then classify the resulting candidates.

The null-regime theorem is therefore used only as a same-type closure theorem [18]. The realized-persistence theorem is invoked only for an already realized standing-bearing continuation and does not turn empty admission into an occupant claim [14].

19.3 Presentation failure and class survival

Let $\text{Fail}_\beta(z)$ be the complete fixed-target failure/undefined/update record of one source presentation. Define

$$\text{Elim}_\beta(q) \iff \forall y \in \widetilde{\mathcal{L}}_\beta(q), y \notin \mathcal{R}_\beta(q). \quad (23)$$

This is a class-level universal statement over the complete raw lift universe, not a first-example test.

Theorem 19.1 ([D1-11] Exact D1 failure and universal non-elimination theorem). *Every generated D1 source point, family/skeleton parameter region, pairwise overlap, and higher-overlap region receives an exact applicability-aware disposition with source and target provenance. Presentation-level exclusion, response-out-of-scope, undefinedness, source failure, and target-update loci may be nonempty. Nevertheless*

$$\boxed{\{q \in Q_{\text{osc}}^{\text{SCIC}} : \text{Elim}_\beta(q)\} = Q_{\text{osc}}^{\text{SCIC}} \setminus \text{im } \rho_{\beta, \text{BB}} = \emptyset.} \quad (24)$$

Nullity, skin authority, same-act repair, hidden governance, spectral overwrite, undeclared source-ontology enlargement, outcome backflow, and target replacement do not create survivors.

Proof. D1-02 gives the complete raw universe and exact partition; D1-03 gives the one-point survivor criterion; and the executed atlas partitions every B15 region before quotienting. The Source-Authorized Process-Compilation Theorem constructs, in the explicitly declared enriched-target extension SOE-D1-1, an endpoint-neutral hyperprocess envelope over every frozen PCR-3 source point. Theorems 8.15, 8.16, and 8.19 prove source coherence, preservation of irreducible higher incidence, current-source deletion naturality, PCR-4 evaluation equality, seven-channel packet equality, and exact recovery of the frozen base source point and SCIC class. Theorem 8.24 then supplies one accepted D1 completion in every base fibre. Hence the universal quantifier in (23) is false for every q . D1-04/D1-05 identify the complement of the base image with the empty-fibre locus, and D1-08 proves full image. Failing intrinsic and overlap strata remain in the exact failure register. $\square$

19.4 Null-regime guardrail

The proof of (24) does not have the form

$$\text{empty residue would be null} \implies \text{residue is nonempty}.$$

Nonemptiness is carried by an independently generated source point, the explicitly admitted hyperprocess-envelope constructor, the source-coherence and deletion-naturality theorems, the exact conservative base-reduct theorem, and the exact D1 completion. AASC then prevents the resulting standing point from being erased by null, skin, repair, or hidden-provenance relabelling.

19.5 Closure and reopening conditions

The universal theorem is reopened by a fixed-target defect such as: failure of the frozen PCR-3 source grammar; failure to declare SOE-D1-1; discovery that the hyperprocess-envelope operation is not conservative; a valid B15 source survivor outside the domain of BB; downstream outcome data entering source formation; flattening of an irreducible higher-arity incidence; failure of source intervention, boundary, matching, vacuum/deformation, continuation, or gluing coherence; failure of current-source deletion naturality; failure of PCR-4 or seven-channel packet equality; inherited D1 endpoint activity in the compiled envelope; an invalid KNM1–5 response; failure of exact base restriction; an overlap or codescent error; or use of AASC null closure as an existence theorem. Later Majorana, heavy-state, cosmological, or symmetry data remain successor targets.

20 Joint-target interface and direct-kinematic closure

20.1 Class-indexed joint-target interface

For every $q \in Q_{\text{osc}}^{\text{SCIC}}$, define

$$\begin{aligned} H_{\beta,q} := & (q, \widetilde{\mathcal{L}}_{\beta}(q), \mathcal{C}_{\beta}, \mathcal{F}_{\beta}(q), \\ & \mathcal{R}_{\beta}(q), \mathcal{L}_{\beta,\text{BB}}(q), \pi_0 \mathcal{L}_{\beta,\text{BB}}(q), \text{ParameterizedMixed}, \\ & \Delta_{\beta,q}^{\text{rel}}, \partial \text{Fail}_{\beta,q}, \text{Prov}_{\beta,q}, \\ & \text{Ov}_{\beta,q}, \text{SOE- D1- 1}, \text{BB}_q). \end{aligned}$$

The interface object contains the candidate-complete raw lift universe, simultaneous D1 constraint bundle, exact complement partition, accepted hyperprocess interval, every intrinsic source-faithful component, parameter descent, source composition, standing separators, failure and update frontiers, overlap incidence, and the source-extension data required by PCR-5E.

At family/skeleton resolution,

$$\mathfrak{H}_{\beta;F,\sigma} = (\widetilde{Z}_{\beta;F,\sigma}^K, \mathcal{C}_{\beta;F,\sigma}, \mathcal{F}_{\beta;F,\sigma}, Z_{\beta;F,\sigma}^K, \Gamma_{\beta;F,\sigma}^K, p_{\beta;F,\sigma}, R_{\beta}, \partial \text{Fail}_{\beta;F,\sigma}, \text{Ov}_{\beta;F,\sigma}, \text{BB}_{F,\sigma}).$$

The interface comprises six family objects, fifteen pairwise bands, forty-two distinct-family higher sectors, and the repeated-label finite-tuple construction.

Theorem 20.1 ([D1-12] Exact populated direct-kinematic handoff theorem). *The class-indexed and family/skeleton records contain all D1 candidate, constraint, standing, source, response, absolute-scale, overlap, separator, failure, update, provenance, and source-extension information required by PCR-5E. They export:*

1. the full base image $\text{im } \rho_{\beta, \text{BB}} = Q_{\text{osc}}^{\text{SCIC}}$;
2. the exact lift groupoid and nonempty residue over every base class;
3. a choice-dependent continuum hyperprocess-envelope subfibre over every base class;
4. the empty class-elimination locus;
5. every pairwise/higher overlap and provenance identification;
6. the explicit *SOE-D1-1* source-ontology extension, its frozen-PCR-3 source basis, and the proof that its base-reduct functor returns the frozen PCR-3 source point and standing class;
7. the complete constituent hypergraph, current-source deletion action, irreducible higher-incidence data, endpoint-neutrality certificate, PCR-4 evaluation equality, and seven-channel standing-packet equality;
8. the non-generative AASC boundary distinguishing source construction from null/skin/repair closure.

PCR-5E therefore need not reconstruct or mutate the D1 quotient.

Proof. D1-01 fixes the target. D1-02 gives the candidate-complete raw universe and exact partition; D1-03 gives the one-point survivor; D1-04 fixes restriction and packet recovery; D1-05 gives the fibres; D1-06/D1-07 give parameter and source descent; D1-08/D1-09 give the full image and universal splitting; D1-10 gives the overlap nerve; and D1-11 gives exact failure provenance and non-elimination. The source-extension, hyperprocess-compilation, deletion-naturality, endpoint-neutrality, and conservative base-reduct records are direct exports of Section 8. $\square$

20.2 Capstone theorem

Theorem 20.2 (Direct Kinematic-Mass Target-Extension Closure Theorem). *Assume the frozen PCR-5B base atlas, the accepted PCR-5C target-extension calculus, and the checksum-verified KNM1–5 target with $\text{D1DataReady} = 1$. Then:*

1. $j_{\beta} : T_{\text{osc}} \hookrightarrow \hat{T}_K^{\text{BB}}$ is a lawful target extension;
2. standing restriction induces

$$R_{\beta, \text{BB}} : \mathcal{N}(\hat{T}_K^{\text{BB}}) \rightarrow \mathcal{N}(T_{\text{osc}}), \quad \rho_{\beta, \text{BB}} = \pi_0(R_{\beta, \text{BB}}),$$

and distinct base SCIC classes do not merge;

3. every base class has a candidate-complete raw lift universe and a total typed constraint partition;
4. AASC closes incoherent null, skin, hidden-governance, outside-governing, and same-act-repair cells, but generates no source candidate and infers no occupant from empty admission;
5. frozen PCR-3 does not contain a universal black-box constructor; PCR-5D1 therefore admits explicitly the enriched-target source-ontology extension *SOE-D1-1*, whose hyperprocess-envelope constructor BB retains every constituent, every primitive incidence, every irreducible higher relation, and all source-constitutive equations;
6. source-kind deletion is lawful on the compiled current source and commutes with compilation, while complete downstream evaluation proves

$$\Phi_4^{\text{ret}}(\text{BB}(H)) = \Phi_4(H), \quad \Xi_{\text{osc}}^{\text{ret}}(\text{BB}(H)) = \Xi_{\text{osc}}(H),$$

including $\Delta_{\text{fr}}^{\text{del}}$;

7. the enriched-source extension has exact frozen base reduct,

$$U_{\beta}^{\text{BB}}(\text{BB}(H)) = H,$$

and $\text{BB}(H)$ contains no inherited absolute beta spectral object before the D1 metric law is attached;

8. after choosing one B15 framed source presentation for each base class, adjoining the one-support D1 metric law to its endpoint-neutral hyperprocess envelope gives an exact KNM1–5 survivor for every $t \in [0, m_K^2)$ and restricts to that base class;

9. therefore

$$\boxed{\text{im } \rho_{\beta, \text{BB}} = Q_{\text{osc}}^{\text{SCIC}}};$$

no base SCIC class is eliminated or selected;

10. for every $q \in Q_{\text{osc}}^{\text{SCIC}}$,

$$\boxed{|\pi_0 \mathcal{L}_{\beta, \text{BB}}(q)| \geq \mathfrak{c}},$$

so every base class is continuum-split and has coarse status `ParameterizedMixed`;

11. the exact enriched quotient is

$$\boxed{Q_{\text{osc}+\beta}^{\text{SCIC}, K, \text{BB}} \cong \bigsqcup_{q \in Q_{\text{osc}}^{\text{SCIC}}} \pi_0 \mathcal{L}_{\beta, \text{BB}}(q)};$$

12. intrinsic endpoint realizations and every typed complement stratum remain source faithful and are never overwritten by the hyperprocess-envelope section;

13. the six family atlases, all pairwise/higher overlaps, failure frontiers, and the PCR-5E handoff are executed over the complete B15 atlas;

14. no class is selected by naturalness, ordering preference, mechanism familiarity, provenance, family label, or finite sampling.

Proof. Items 1–2 are D1-01/D1-04. Items 3–7 are D1-02/D1-03 and Theorems 8.6, 8.9, 8.15, 8.16, 8.17, 8.18, and 8.19. Item 8 is Theorem 8.22 together with a choice of one B15 frame over each q . Item 9 is D1-08 and Theorem 8.24; item 10 is D1-09; item 11 is D1-05; item 12 is the complete source-faithful residue calculus; and items 13–14 are D1-10 through D1-12 plus the frozen no-backflow rules. $\square$

Strongest D1 endpoint

PCR-5D1 closes the direct-kinematic target at exact groupoid/moduli resolution:

$$\boxed{\text{im } \rho_{\beta, \text{BB}} = Q_{\text{osc}}^{\text{SCIC}}, \quad Q_{\text{osc}+\beta}^{\text{SCIC}, K, \text{BB}} \cong \bigsqcup_{q \in Q_{\text{osc}}^{\text{SCIC}}} \pi_0 \mathcal{L}_{\beta, \text{BB}}(q), \quad |\pi_0 \mathcal{L}_{\beta, \text{BB}}(q)| \geq \mathfrak{c} \ \forall q.}$$

The KATRIN upper-limit target eliminates no oscillation SCIC class but refines every one into at least continuum many standing-distinct direct-kinematic realizations.

20.3 Physical meaning

The proof has four distinct layers. Frozen PCR-3/B15 generates the original source-complete base atlas. PCR-5D1 declares a narrowly scoped hyperprocess-envelope extension because generic process compilation is not already an operation of frozen PCR-3. The complete D1 and AASC constraints then partition the enriched candidate universe. Finally, the source-authorized compilation, current-source deletion naturality, higher-incidence preservation, endpoint neutrality, exact base reduct, and exact KATRIN response provide one accepted residue in every base fibre. The null-regime and realized-persistence theorems classify incoherent closure routes; they do not create the residue.

Intrinsic mass-bearing source presentations retain their complete source-derived spectra and can pass or fail individually. The hyperprocess envelope is a distinct lawful presentation in the explicit enriched-target extension. It retains the full constituent architecture while exposing an endpoint-neutral external process; it supplies a choice-dependent accepted metric interval without overwriting intrinsic physics or encoding a solved standing core into source formation.

20.4 Subsequent neutrino target extensions

PCR-5D2 adds Majorana/LNV and neutrinoless-double-beta-decay standing; PCR-5D3 heavy-state, nonunitarity, sterile/right-handed-sector, and charged-lepton-flavour data; PCR-5D4 cosmological and astrophysical transport; and PCR-5D5 local, global, accidental, gauged, and remnant lepton-symmetry structure. Each successor target may further eliminate or split D1 fibres but cannot retroactively alter D1 membership or equivalence.

20.5 Conditions for failure

The capstone is reopened by:

- a proof that SOE-D1-1 is not a lawful enriched-source extension or that its declared base reduct fails;
- a valid B15 source survivor outside the domain of BB;
- downstream interface, failure, standing, survivor, or D1 values entering source formation;
- flattening or omission of an irreducible higher-arity source incidence;
- failure of current-source deletion naturality;
- failure of PCR-4 evaluation equality or one of the seven PCR-5B packet channels;
- an inherited beta-endpoint spectral object in the hyperprocess envelope;
- use of AASC null closure as an existence theorem;
- a lawful D1 completion omitted from the raw universe;
- a missing KNM1–5 record, response term, covariance, calibration, or statistical convention;
- failure of exact base restriction or standing separation of unequal metric points;
- an overlap or codescent identification that merges unequal enriched standing cores;
- a later target datum used to alter D1 membership retroactively.

20.6 Conclusion

The complete KNM1–5 target and its comparison class determine a carrier-neutral spectral response, a class-indexed lift universe, an exact constraint partition, source-faithful intrinsic endpoint components, a conservative hyperprocess extension, current-source deletion, absolute-scale lift fibres, their intersections, and the interface required for joint neutrino closure. The full-image and continuum-splitting results follow from the explicitly declared source operation together with the simultaneous constraint calculus; they do not rely on a generative reading of AASC or on identifying a downstream standing core with a source object.

A Notation and type hierarchy

Table 3: Principal PCR-5D1 notation.

Symbol	Meaning
T_{osc}	Frozen neutrino-oscillation target
$\hat{T}_K^{\text{BB}}$	Audited KATRIN KNM1–5 direct endpoint target
$\mathcal{N}(T)$	Concrete standing-core groupoid at target T
$Q(T) = \pi_0 \mathcal{N}(T)$	SCIC quotient at target T
R_β	Standing-core restriction from $\hat{T}_K^{\text{BB}}$ to T_{osc}
ρ_β	Restriction on connected components
$\mathcal{L}_\beta^K(q)$	Exact direct-kinematic lift groupoid over base class q
μ_e	Electron-boundary mass-squared spectral measure
m_β^2	Effective unresolved electron-neutrino mass-squared moment
$\mathcal{D}_K$	Audited campaign data record
$\mathcal{I}_K$	Audited response/nuisance input record
Resp_K	Frozen KATRIN response object
$\mathcal{L}_K$	Frozen likelihood/statistical object
$\mathcal{A}_K$	Frozen acceptance/confidence object
$\Delta_\beta^{\text{rel}}$	Enriched-only standing separator family
MinSuff_β	Minimal no-repair sufficient source subsets

A.1 Identity hierarchy

The paper uses four equivalence levels:

1. source-presentation equivalence inside the PCR-3 candidate groupoid;
2. base SCIC equivalence at T_{osc} ;
3. enriched response/presentation equivalence at $\hat{T}_K^{\text{BB}}$;
4. enriched SCIC equivalence after complete standing-core descent.

Equality at one level does not imply equality at a finer level. In particular, equal oscillation packets do not imply equal direct endpoint responses, and equal m_β^2 does not imply equal enriched standing cores

outside the unresolved branch.

A.2 Target, inference, and physical domains

The physical spectral domain is supported on nonnegative mass-squared. The estimator domain may be larger. The response/nuisance domain contains calibrated apparatus coordinates. The base carrier domain contains source-complete neutrino carrier presentations. These domains are linked by typed maps rather than identified.

B Audited KATRIN public data instance

B.1 Integrity and acquisition

The supplementary source package contains the authoritative JSON byte streams under `dependencies/katrin`. Their official and local checksums are given in the data-provenance tables. The field-level audit verifies 2,383 structural conditions without a hard failure.

B.2 Campaign details

The seven campaign blocks differ in scan length, detector grouping, response configuration, and Penning-timing representation. The target does not collapse these blocks into one averaged spectrum before response evaluation.

Table 4: Complete campaign summary from the public record.

Campaign	Points	Width	Cells	Counts	Live days	Penning form
KNM1	27	1	27	2,029,854	21.736	scalar
KNM2	28	1	28	4,314,055	28.927	scalar
KNM3-SAP	28	14	392	1,069,831	9.173	scalar
KNM3-NAP	28	1	28	1,426,505	9.339	scalar
KNM4-NOM	28	14	392	5,637,508	34.783	three-component
KNM4-OPT	25	14	350	4,582,387	17.990	scalar zero-valued timing
KNM5	28	14	392	16,652,705	51.067	scalar zero-valued timing

B.3 Matrix audit

Nine covariance/correlation objects were checked for dimensions, finite entries, symmetry to released precision, diagonal semantics, correlation bounds, and positive semidefiniteness of the symmetrized matrix. The gas-density covariance has relative finite-precision asymmetry approximately 2.79×10^{-5} ; this is retained as a released-data fact.

B.4 Published-data conventions and qualifications

- Three transmission-broadening uncertainty entries are explicit nulls and are not converted to zero.
- Potential-drift broadening-squared includes signed fit inputs and is not clipped.

- Background-overdispersion pixel metadata groups twelve values into three listed pixel-group assignments.
- No attempt is made to infer missing uncertainty semantics from neighboring fields.

C Endpoint-spectrum and response derivations

C.1 Discrete spectral specialization

For a finite discrete measure $\mu_e = \sum_a w_a \delta_{m_a^2}$, the target phase-space factor is

$$\int \varepsilon \sqrt{\varepsilon^2 - m^2} \Theta(\varepsilon - m) d\mu_e(m^2) = \sum_a w_a \varepsilon \sqrt{\varepsilon^2 - m_a^2} \Theta(\varepsilon - m_a).$$

This recovers the conventional incoherent endpoint sum.

C.2 Molecular final states

The molecular final-state distribution replaces $E_0 - E$ by $E_0 - V_f - E$ and averages over P_f . The updated final-state uncertainty treatment is part of the target model rather than a carrier selector [19].

C.3 Integral response

The MAC-E filter measures an integral spectrum. For detector group p and retarding potential U , the signal rate is

$$\dot{N}_p^{\text{sig}}(U) = \int dE S_\beta(E) R_p(E, U).$$

The response contains transmission, energy loss, field/potential inhomogeneity, source scattering, detector acceptance, and campaign configuration. The predicted count at scan step k multiplies the rate by live time and relative efficiency and adds the background contribution.

C.4 Likelihood factorization

The target likelihood can be represented schematically as

$$-2 \log \mathcal{L}_K = -2 \sum_{c,p,k} \log \text{Pois}(N_{c,p,k} \mid \lambda_{c,p,k}) + \mathcal{Q}_K(\eta),$$

where $\mathcal{Q}_K$ collects the frozen nuisance/covariance constraints or their exact collaboration-approved equivalent. This schematic form is used only to type the dependence structure. The paper imports the published inference rather than claiming a new numerical fit.

C.5 Response-equivalence criterion

Two spectral objects are response-equivalent only when the full vector Λ_K , acceptance status, matching, boundary, and failure data agree after lawful nuisance/calibration transport. Equal first moments do not suffice in a regime sensitive to higher spectral structure.

D Standard unresolved branch and scale intervals

D.1 General affine formula

Let o label any retained relative-shape branch and let w be a unit-normalized electron-weight vector. With $m_i^2 = t + \delta_i(o)$,

$$m_\beta^2 = t + c(o, w), \quad c(o, w) = \sum_i w_i \delta_i(o).$$

The physical lower bound is

$$t \geq t_{\min}(o) = \max_i \{-\delta_i(o), 0\}$$

when t is chosen so that every m_i^2 is nonnegative.

D.2 Projected KATRIN interval

The scalar upper-limit envelope produces

$$\mathcal{I}_K(o, w) = [t_{\min}(o), M_K^2 - c(o, w)]$$

when the upper endpoint exceeds the lower endpoint. The open upper endpoint reflects the strict reported inequality; using a closed confidence boundary would replace it by the corresponding closed endpoint without changing non-singletonhood when the interval has positive length.

D.3 No ordering preference

The formula applies separately to every branch. A numerical branch comparison requires frozen oscillation shape and weight inputs. PCR-5D1 does not import current global-fit numbers as hidden premises. The theorem therefore remains symbolic at the base-atlas level.

D.4 Zero-lightest and positive-floor subloci

The zero-lightest boundary is the locus where $t = t_{\min}(o)$ and at least one $m_i^2 = 0$. The positive-floor locus has $t > t_{\min}(o)$. Both can lie inside the scalar KATRIN interval. They are separate enriched SCIC classes only if the complete endpoint standing distinguishes them.

E General spectral measures and response validity

E.1 Finite resolved spectra

A finite spectrum with separations large enough to produce target-relative kinks cannot be reduced automatically to m_β^2 . The correct comparison uses the full discrete spectral measure in the differential spectrum and response integral.

E.2 Continuous and functional spectra

A continuous component may arise from an enlarged state space, environmental coupling, or another source architecture admitted by PCR-3. It is represented by an absolutely continuous or mixed part of μ_e . Such a candidate is evaluated only when the response and likelihood are defined on that spectral class.

E.3 ResponseOutOfScope versus failure

Proposition E.1. If a source-complete D1 candidate has a valid base reduct and target-typed electron-boundary interaction but the frozen KNM1–5 response model does not define its predicted count vector,

then the correct status is `ResponseOutOfScope` or `TargetUpdate`, not `AcceptanceFail`.

Proof. Acceptance requires a defined response and likelihood. Undefinedness of the comparison map cannot establish physical mismatch. $\square$

E.4 Moment degeneracy

Distinct measures can have the same first moment:

$$\int m^2 d\mu_1 = \int m^2 d\mu_2$$

while differing in higher moments or support. The complete response, not the first moment alone, decides whether the target identifies them.

F D1 source grammar and lift-groupoid data

F.1 Complete endpoint-potential source diagrams

For a PCR-5B source presentation x , let $\text{EPot}_\beta(x)$ be the source-constitutive closure of every inherited object, incidence, law, mass map, matching map, boundary attachment, constituent/profile datum, hidden projection, environmental or topological support, and higher relation capable of participating in the tritium electron-boundary endpoint after the D1 target is activated. The diagram is formed before any D1 response is evaluated.

Each endpoint-potential object receives exactly one status

$$\text{Direct}, \quad \text{Routed}, \quad \text{SourceNull}, \quad \text{TargetUpdate}.$$

A Routed object remains physically the same inherited contribution: its route may factor the endpoint incidence but must preserve the source-derived spectral measure and boundary weight. A spectral replacement not derived from a source constitutive relation is not routing.

F.2 Source-faithful completion datum

A raw D1 datum over a base presentation θ_0 has the form

$$\begin{aligned} \text{RawRec}_\beta(\theta_0, \eta) = & (\theta_0, \phi_0, \text{EPot}_\beta(\theta_0), \text{Act}_\beta, \{\varrho_{\beta,a}\}_{a \in R(\theta_0)}, \\ & \{\mu_{\beta,a}^{\text{src}}\}_{a \in D(\theta_0) \cup R(\theta_0)}, \eta, \mu_\beta^{\text{new}}(\eta), \mu_\beta^{\text{tot}}, \\ & \text{Own}_e, \text{Match}_\beta, \eta_K, \Lambda_K, \mathcal{L}_K, \text{Applicability}, \text{Status}). \end{aligned}$$

Here ϕ_0 frames the exact reduct to the PCR-5B provenance presentation, and

$$\mu_\beta^{\text{tot}} = \sum_{a \in D} \mu_{\beta,a}^{\text{src}} + \sum_{a \in R} (\varrho_{\beta,a}) * \mu_{\beta,a}^{\text{src}} + \mu_\beta^{\text{new}}(\eta).$$

The datum is accepted only when every endpoint-potential path is dispositioned, every route is spectrally preserving, one common source point realizes all source and target relations, and the complete frozen KNM1–5 evaluator accepts the resulting response.

F.3 Lawful source arrows

Lawful arrows include typed base-source isomorphisms, source-equational automorphisms, boundary-preserving basis/gauge changes, spectral-preserving route-frame isomorphisms, and response-calibration maps already licensed by the target. They preserve the complete endpoint-activity partition, source-derived spectral content, enriched standing response, and same-owner beta matching. A change of physical endpoint support, inherited source equations, or frozen target semantics belongs to another scientific point or a TargetUpdate branch rather than to a redescription.

G Overlap groupoids and higher coherence

G.1 Fixed-core formula

For a concrete enriched core n , a pairwise overlap can be written

$$\left(\widetilde{\text{Prov}}_{\beta,a}(n) \times \widetilde{\text{Prov}}_{\beta,b}(n) \right) // \text{Aut}(n).$$

The diagonal automorphism action changes frames, not the common standing core. Equality requires equality of the frozen base core together with the complete endpoint-active spectral object, response-equivalence class, boundary/matching data, parameter-descent class, and failure and update data.

G.2 First-difference separator

When two survivor presentations do not overlap, the complement datum identifies the first unequal standing layer in the order

$$\Xi_{\text{osc}} < K_{\beta}^{\text{act}} < K_{\beta}^{\text{spec}} < K_{\beta}^{\text{resp}} < K_{\beta}^{\text{e-att}} < K_{\beta}^{\text{match}} < K_{\beta}^{\text{fail}}.$$

This is a reporting order, not a ranking of physical importance. A route that changes spectral support is recorded before response equality is considered because it is already source/target invalid.

G.3 Higher Čech nerve

The higher nerve is built from repeated homotopy fibre products over $\mathcal{N}(\widehat{T}_K^{\text{BB}})$. Face maps forget a presentation, degeneracies repeat one with an identity frame, and cocycles follow from composition of complete-core isomorphisms. Each higher overlap therefore carries one coherent enriched standing core while retaining every source presentation in framed provenance.

G.4 Connected-subgroupoid fibre proof

Let $R_{\beta} : \mathcal{N}(\widehat{T}_K^{\text{BB}}) \rightarrow \mathcal{N}(T_{\text{osc}})$ be standing-core restriction. For a connected base subgroupoid $\mathcal{N}(T_{\text{osc}})[q]$, the homotopy pullback $\mathcal{L}_{\beta,\text{BB}}(q)$ contains precisely accepted enriched cores equipped with a base frame into q . Passing to connected components yields

$$\pi_0 \mathcal{L}_{\beta,\text{BB}}(q) \cong \rho_{\beta,\text{BB}}^{-1}(q).$$

An empty accepted completion object for one source presentation is not enough to empty this homotopy fibre because other framed provenance presentations of the same base core may remain.

G.5 Global assembly

The coarse disjoint-union formula follows because $\rho_{\beta, \text{BB}}$ is a function. At groupoid level, the framed lift pseudofunctor admits the PCR-5C Grothendieck reconstruction of $\mathcal{N}(\hat{T}_K^{\text{BB}})$. The family atlas is a presentation cover of that enriched core groupoid, and codescent glues only already classified equal cores. The accepted base image is the essential image of restriction, equivalently the codescent of the accepted chart projections.

H Adversarial tests and reopening conditions

The following adversarial transformations test consequences of the theorem chain. They do not replace the semantic proofs of source completeness, target typing, standing identity, or fibre exhaustion. Each transformation must fail for the stated mathematical reason.

Table 5: Principal adversarial transformations of the source-extension theorem.

Transformation	Required disposition
Attribute the generic BB constructor to frozen PCR-3 saturation without declaring a source-ontology extension	Reject as FrozenPCR3Attribution
Remove SOE-D1-1 while retaining the hyperprocess-envelope witness	Reject as SourceOntologyExtensionUndeclared
Allow the constructor signature to read a PCR-4 pass/fail result, PCR-5B standing core, packet value, survivor label, failure record, or KATRIN outcome	Reject as OutcomeBackflow
Define $\text{BB}(H)$ by a lookup table returning the desired seven-channel packet	Reject as downstream witness formation
Flatten an irreducible fusion, inflow, common-environment, collective, or many-body hyperedge into pairwise/pathwise operations	Reopen source coherence and higher-incidence preservation
Omit one primitive source node, incidence, parameter, boundary, matching arrow, vacuum datum, extension cell, or higher-coherence relation from $\mathbb{P}(H)$	Reopen the Source-Authorized Process-Compilation Theorem
Alter one source parameter while compiling	Reject as source mutation rather than conservative presentation
Evaluate the frozen deletion signature on the provenance history of H rather than by an intervention on the compiled current source	Reject as CurrentSourceDeletionFailure
Force $\text{Del}_S^{\text{BB}}(\text{BB}(H)) \not\cong \text{BB}(\text{Del}_S H)$	Reopen deletion naturality and D1-04
Retune a retained parameter, matching datum, vacuum branch, boundary, gluing map, or continuation after deletion	Reject as no-repair violation
Change one PCR-4 evaluator input under base restriction	Reopen evaluator equality

Transformation	Required disposition
Change one of $T, R, \Delta_{\text{fr}}^{\text{del}}, C, M, B, F$ after complete evaluation	Reopen retained-target evaluation, exact base reduct, and D1-04
Let the hyperprocess envelope contain an absolute beta spectral sort before D1 completion	Reject as EndpointNeutralityFailure
Let the base-reduct functor return only a standing packet rather than the exact frozen source point	Reject as source-reduct failure
Add, delete, split, or merge an oscillation SCIC class under the source extension	Reject as ConservativeQuotientFailure
Attach the one-support metric law to a second primitive carrier node	Reject by no-side-carrier discipline
Overwrite an intrinsic multi-support endpoint spectrum by the one-support law	Reject as spectral overwrite/target update
Treat the hyperprocess-envelope branch as the same PCR-3 source presentation as an intrinsic mass-bearing branch	Reject; they are distinct provenance objects before terminal standing descent
Use an intrinsic source failure to eliminate its entire base SCIC class without exhausting all framed provenance	Reject by class-level fibre exhaustion
Infer nonempty residue solely because an empty residue would be null	Reject; AASC is non-generative
Insert a new candidate after observing KNM1–5 failure	Reject as post-census candidate insertion/same-act repair
Change the weak source interaction, tritium source semantics, detector response, or confidence construction while calling the result the same D1 target	Target update
Identify δ_t and $\delta_{t'}$ for $t \neq t'$	Reject by D1 absolute-support standing
Demand a canonical representative-independent section over each base class	Reject as unnecessary overclaim; a choice-dependent section suffices for surjectivity
Use family name, naturalness, field count, mechanism familiarity, or provenance as a carrier selector	Reject by SCIC/no-second-classifier
Omit one KATRIN campaign, covariance, calibration, nuisance, response, or likelihood coordinate	Reopen D1-01
Replace the complete symbolic moduli atlas by finitely many examples	Reject finite sampling as exhaustion
Omit one family-pair or higher-overlap band	Reopen D1-10
Promote a CSV row, validator PASS flag, or hash identity into semantic proof	Reject certification substitution
Mutate D1 membership or equivalence during PCR-5E	Reject no-backflow violation

The supplementary test matrix extends these transformations across all six families, the marked classes, pairwise and higher intersections, source-extension typing, current-source deletion, response, failure, and joint-target loci. A scientific reopening must preserve the fixed target and exhibit a named source, constraint, restriction, response, fibre, overlap, or provenance implication that fails.

I Sources, data provenance, and reproducibility

I.1 Supplementary source package

The supplementary package contains:

- the complete LaTeX manuscript and bibliography;
- the approved detailed theorem skeleton;
- the original audited KATRIN JSON files and source papers;
- the frozen PCR-3, PCR-5B, and PCR-5C dependencies used by the source-extension and target-extension proofs;
- the version-locked null-regime and realized-persistence closure papers;
- campaign, field, matrix, target-map, theorem-dependency, candidate-universe, constraint-residue, source-extension, hyperprocess, deletion, quotient, overlap, adversarial-test, closure-criterion, and joint-target tables;
- deterministic build and consistency-check scripts;
- a complete SHA-256 manifest.

I.2 Reproducibility boundary

Checksums establish file identity. Structural checks establish the declared data layout. Matrix checks verify algebraic properties of released arrays, while LaTeX and PDF checks establish reproducible typesetting. None of these proves candidate completeness, enriched-source-extension validity and exact base-reduct preservation, source coherence, response standing, SCIC identity, fibre exhaustion, or physical realization. Those are established by the semantic arguments in the manuscript.

In particular, the supplementary checks do not prove:

- that AASC generates a candidate;
- that frozen PCR-3 already contained a universal black-box/process compiler;
- that SOE-D1-1 is conservative;
- that the hyperprocess envelope preserves every irreducible higher incidence;
- that current-source deletion is natural;
- that the exact KNM1–5 response supplies a survivor;
- that every base fibre is nonempty or continuum split;
- uniqueness or representative-independence of the chosen section.

I.3 AASC closure dependencies

The structural null-regime theorem is used only as a closure theorem: total vacuity cannot occupy a same-type positive admissibility role. The realized-persistence theorem is used only after an already realized standing-bearing continuation regime has been fixed; it proves non-vacuous reconstruction structure and explicitly does not infer an occupant from an empty evaluated fragment. These dependencies therefore constrain the classification of generated D1 candidates without acting as source generators.

I.4 Source-theory boundary

Frozen PCR-3 supplies finite source-coherent typed hypergraphs, law/process objects, derived objects, finite hybrids, and irreducible higher incidence. It also states that a genuinely new source sort, operation, or constitutive equation changes the source ontology. PCR-5D1 therefore does not attribute the generic envelope constructor to PCR-3 saturation. It declares SOE-D1-1 explicitly and proves its source coherence, current-source deletion naturality, endpoint neutrality, and conservative base quotient inside the D1 enriched comparison class.

I.5 Direct dependency map

Source	Direct use in PCR-5D1
PCR-3 source-first exhaustion	Frozen base source ontology; finite hypergraph/higher-incidence discipline; explicit boundary on new source operations; no outcome backflow
PCR-5B publication final	Exact base SCIC atlas, B15 provenance, separators, deletion packet, and overlaps
PCR-5C publication final	Lawful extension, restriction, lift fibres, no merger, composition, and failure provenance
PCR-4	One-point simultaneous survival, applicability, failure, and evaluator input/output discipline
PCR-5A/SCIC/ATS	Frozen standing packet, deletion discipline, unique terminal carrier identity, and no second classifier
Null-regime closure	Exclusion of null-but-standing and same-type vacuous closure; no existence inference
Realized-persistence non-vacuity	Non-null continuation-sensitive structure of already realized standing cores; no occupant inference from empty admission
KATRIN 259-day result	Published estimator, upper limit, target campaign, and systematic architecture
KATRIN response/method papers	Differential spectrum, apparatus response, nuisance, calibration, and statistical semantics
KATRIN calibration papers	Final-state, shifted-plane field, and source-potential target coordinates
Zenodo KNM1–5 release	Authoritative measured-data and model-input byte streams

J Atlas data and the joint-target interface

J.1 Family-wise lift objects

The enriched atlas is indexed by every PCR-5B family/skeleton chart and its full symbolic parameter domain. Each chart contains the raw D1 lift object, the simultaneous constraint partition, the universal hyperprocess residue defined by SOE-D1-1, and the intrinsic source-faithful endpoint components.

Table 7: Hyperprocess and intrinsic D1 family data.

Family	Base domain	Universal residue	Complete stand- ing object
LAW	every $\sigma \in \Sigma_{\text{LAW}}^{\text{gram}}$	$\text{BB}(H) \times J_{\beta}^{\text{BB}}$ in SOE-D1-1	$Z_{\beta;\text{LAW},\sigma}^K$
LOCAL	every $\sigma \in \Sigma_{\text{LOCAL}}^{\text{gram}}$	$\text{BB}(H) \times J_{\beta}^{\text{BB}}$ in SOE-D1-1	$Z_{\beta;\text{LOCAL},\sigma}^K$
HID	every $\sigma \in \Sigma_{\text{HID}}^{\text{gram}}$	$\text{BB}(H) \times J_{\beta}^{\text{BB}}$ in SOE-D1-1	$Z_{\beta;\text{HID},\sigma}^K$
COMP	every $\sigma \in \Sigma_{\text{COMP}}^{\text{gram}}$	$\text{BB}(H) \times J_{\beta}^{\text{BB}}$ in SOE-D1-1	$Z_{\beta;\text{COMP},\sigma}^K$
SOL	every $\sigma \in \Sigma_{\text{SOL}}^{\text{gram}}$	$\text{BB}(H) \times J_{\beta}^{\text{BB}}$ in SOE-D1-1	$Z_{\beta;\text{SOL},\sigma}^K$
HYB	every $\sigma \in \Sigma_{\text{HYB}}^{\text{gram}}$	$\text{BB}(H) \times J_{\beta}^{\text{BB}}$ in SOE-D1-1	$Z_{\beta;\text{HYB},\sigma}^K$

A hyperprocess point is formed from a framed B15 source presentation and retains every constituent port, primitive incidence, source parameter, boundary and matching datum, representation/charge/anomaly entry, vacuum/deformation branch, continuation, gluing relation, applicability value, failure datum, and current-source deletion action. Its standing packet is evaluated only after source formation. Intrinsic components independently retain their source-derived endpoint spectra and exact response dispositions.

J.2 Source extension and conservative base descent

For each family the atlas specifies:

1. the SOE-D1-1 source signature and its frozen PCR-3 basis;
2. the hyperprocess constructor and source equations;
3. preservation of irreducible higher incidence;
4. naturality of deletion on the current enriched source;
5. endpoint neutrality before the D1 metric attachment;
6. equality of the retained PCR-4 evaluation and the seven-channel PCR-5B packet;
7. the exact base reduct and standing-quotient map.

These data make the enlarged D1 source comparison class part of the target rather than an implicit auxiliary convention.

J.3 Constraint residues

For every family/skeleton chart the central object is

$$(\tilde{Z}_{\beta;F,\sigma}, \mathcal{C}_{\beta;F,\sigma}, \mathcal{F}_{\beta;F,\sigma}, Z_{\beta;F,\sigma}^K),$$

where $\tilde{Z}_{\beta;F,\sigma}$ is the complete raw lift object relative to $\hat{T}_K^{\text{BB}}$, $\mathcal{C}_{\beta;F,\sigma}$ is the simultaneous D1 constraint bundle, $\mathcal{F}_{\beta;F,\sigma}$ is its typed complement, and $Z_{\beta;F,\sigma}^K$ is the standing-positive residue. AASC closes only the null, skin-authority, hidden-governance, outside-but-governing, and same-act-repair cells; positive source points are supplied by the declared source theory and then evaluated.

J.4 Marked calibrations

The classes $q_W, q_{3N}, q_{\text{red}}, q_{\text{syn}}$ provide four calibration points. Their hyperprocess presentations recover the corresponding seven-channel base packets and the current-source deletion values

$$(0, 1), \quad (1, 0), \quad (1, 1), \quad (0, 0),$$

respectively. Each class contains the continuum D1 family indexed by J_{β}^{BB} , while the original Weinberg, active-singlet, redundant, and synergistic presentations remain in their source-derived endpoint branches.

J.5 Pairwise and higher intersections

For each of the fifteen unordered family pairs, the atlas contains the exact standing-core equalizer over every corresponding skeleton pair. Hyperprocess intersections require a common base core and common t ; intrinsic intersections use the complete enriched standing signature. The forty-two distinct-family higher sectors and repeated-label finite tuples are obtained by the corresponding iterated homotopy fibre products.

J.6 Interface for PCR-5E

The joint-target interface consists of sixty-seven class-, family-, pair-, and higher-indexed objects. Together they retain the raw lift universe, constraint and failure partition, full base image, **ParameterizedMixed** fibre type, target-complete identifier, source-extension data, hyperprocess and intrinsic moduli, base restriction, relative separators, framed provenance, failure/update boundary, and overlap incidence. The interface carries

$$\text{im } \rho_{\beta, \text{BB}} = Q_{\text{osc}}^{\text{SCIC}}, \quad |\pi_0 \mathcal{L}_{\beta, \text{BB}}(q)| \geq \mathfrak{c} \quad \forall q,$$

while preserving every presentation-level failure and every noncanonical endpoint branch.

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